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1. Introduction

This document models connected coffee machines that send `telemetry` to a `central cloud site`, receive OTA firmware updates, and produce operational and audit logging evidence. Authored functional architecture remains stable while runtime, code, and verification layers are inferred from infrastructure, source, and test artifacts.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`)	stable	Stable authored capability boundaries and responsibility ownership.	7	7	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Communication View (`VIEW-COMMUNICATION`)	in-review	Communication handoffs are mapped; protocol detail remains partially inferred.	7	4	3	medium	Counts a unit as covered when direct inferred runtime evidence exists.
Deployment View (`VIEW-DEPLOYMENT`)	in-review	Deployment/runtime mapping is present with evolving control-plane evidence depth.	7	4	3	medium	Counts a unit as covered when direct inferred runtime evidence exists.
Security View (`VIEW-SECURITY`)	draft	Core threats are modeled while additional control evidence is being expanded.	7	7	1	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (`VIEW-TRACEABILITY`)	in-review	Requirement mappings exist; verification and evidence coverage continue to be hardened.	7	7	3	high	Counts a unit as covered when direct inferred runtime or code evidence exists.
State Lifecycle View (`VIEW-STATE-LIFECYCLE`)	in-review	OTA lifecycle transitions and guards are authored for deterministic operational progression.	0	0	0	n/a	Lifecycle heuristic is currently basic; treat authoredStatus as primary signal.
Interaction Flow View (`VIEW-INTERACTION-FLOW`)	in-review	OTA rollout interaction sequence is explicitly modeled from operator intent to rollout reporting.	5	5	0	high	Flow heuristic is step/branch-oriented; treat authoredStatus plus projection completeness as the primary signal.

4. Terms and Definitions

Term	Definition
`actor` (EM-ACTOR)	A person or operational role that interacts with functional units.
`attack vector` (EM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
`audit log record` (DATA-AUDIT-LOG-RECORD)	Immutable log record for security and compliance analysis. Aliases:audit records, logging records
`audit logging is enabled` (FEAT-AUDIT-LOGGING)	Feature flag enabling audit-grade log persistence. Aliases:audit enabled
`authored mapping` (EM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
`barista` (ACT-BARISTA)	Operates coffee machines and observes local machine health. Aliases:store operator
`brew cycle completed event is received` (EVT-BREW-CYCLE-COMPLETED)	Event raised when a coffee machine completes a brew cycle. Aliases:brew completed, brew cycle event
`brew telemetry` (DATA-BREW-TELEMETRY)	Machine telemetry including brew duration, temperature, and status metadata. Aliases:machine status, machine telemetry, telemetry, telemetry payloads, telemetry records
`campaign id` (DATA-CAMPAIGN-ID)	Stable OTA campaign identifier used to correlate rollout and audit records. Aliases:campaign identifier, campaign ids
`cloud control` (FG-CLOUD-CONTROL)	Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Aliases:central cloud site
`cloud ingest endpoint is unavailable` (EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE)	Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion. Aliases:endpoint outage, ingest endpoint unavailable
`cloud logging service` (REF-CLOUD-LOGGING-SERVICE)	Managed log and metrics service used for fleet observability. Aliases:logging service
`cloud runtime operations` (FU-CLOUD-RUNTIME-OPERATIONS)	Operates cloud runtime deployment and scaling controls. Aliases:runtime operations
`cloud runtime sdk` (REF-CLOUD-RUNTIME-SDK)	SDK used by cloud units for runtime integrations. Aliases:cloud sdk
`code element` (EM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
`coffee fleet system` (SYS-COFFEE-FLEET-SYSTEM)	Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture. Aliases:the coffee fleet system
`coffee machine fleet operation is active` (STATE-FLEET-OPERATION-ACTIVE)	State where machines and central cloud are actively serving operations. Aliases:fleet operation active

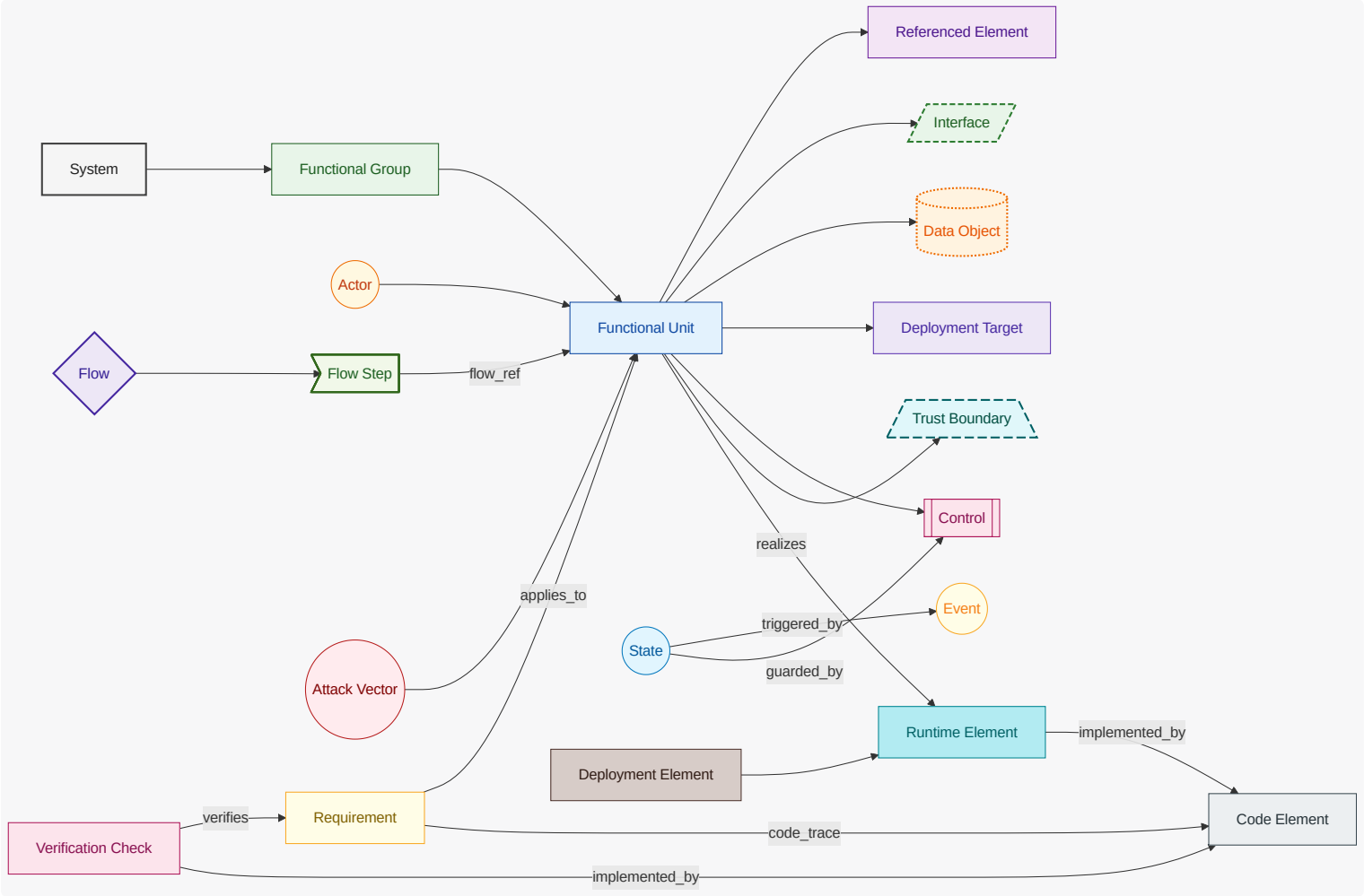
Term	Definition
`control` (EM-CONTROL)	An authored security, policy, or operational control used to constrain behavior and reduce risk.
`data object` (EM-DATA-OBJECT)	An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.
`deployment target` (EM-DEPLOYMENT-TARGET)	An authored deployment destination such as environment, cluster, namespace, or registry scope.
`device identity and secrets` (FU-DEVICE-IDENTITY-SECRETS)	Manages device identity material and secure configuration. Aliases:device secrets
`event` (EM-EVENT)	An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.
`firmware bundle store` (REF-FIRMWARE-BUNDLE-STORE)	Object store containing signed OTA firmware bundles. Aliases:firmware store
`firmware signature is valid` (COND-FIRMWARE-SIGNATURE-VALID)	Condition indicating firmware signature verification succeeded. Aliases:signature valid
`firmware signer service` (REF-FIRMWARE-SIGNER-SERVICE)	Service that signs firmware artifacts before OTA rollout. Aliases:signer service
`firmware validation failed event is received` (EVT-FIRMWARE-VALIDATION-FAILED)	Event raised when signature/hash validation fails on firmware. Aliases:validation failed
`fleet ingestion api` (FU-FLEET-INGESTION-API)	Ingests telemetry and machine status from edge devices. Aliases:ingestion api
`fleet observability reporting` (FU-FLEET-OBSERVABILITY-REPORTING)	Aggregates telemetry and update outcomes into operational and audit signals. Aliases:observability reporting
`fleet operator` (ACT-FLEET-OPERATOR)	Owns OTA rollout campaigns and fleet-level reliability. Aliases:fleet admin
`flow` (EM-FLOW)	An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.
`flow step` (EM-FLOW-STEP)	A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.
`functional group` (EM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
`functional unit` (EM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
`inferred mapping` (EM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.

Term	Definition
`interface` (EM-INTERFACE)	An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.
`iot ingest endpoint` (REF-IOT-INGEST-ENDPOINT)	Cloud endpoint for receiving telemetry and status from devices. Aliases:iot endpoint
`log tampering` (AV-LOG-TAMPERING)	Attempts to suppress or alter security/operational log records. Aliases:audit tampering
`machine edge` (FG-MACHINE-EDGE)	Capability area for device-local telemetry and OTA execution behavior. Aliases:coffee machine edge
`machine identity` (DATA-MACHINE-IDENTITY)	Device identity material used for authenticated publish, ingest, and audit attribution. Aliases:device identity, machine identities
`machine telemetry collection` (FU-MACHINE-TELEMETRY-COLLECTION)	Collects brew/device telemetry and publishes cloud-bound payloads. Aliases:telemetry collection
`malicious firmware bundle` (AV-MALICIOUS-FIRMWARE-BUNDLE)	Tampered firmware payload attempts to execute on devices. Aliases:firmware tampering
`mqtt device sdk` (REF-MQTT-DEVICE-SDK)	SDK used by edge components for telemetry and command transport. Aliases:mqtt sdk
`normal mode is active` (MODE-NORMAL)	Default operation mode for telemetry and OTA behavior. Aliases:normal mode, normal operation mode
`ota campaign is enabled` (FEAT-OTA-CAMPAIGN)	Feature flag enabling centralized OTA rollout behavior. Aliases:ota enabled, ota rollout
`ota campaign policy` (DATA-OTA-CAMPAIGN-POLICY)	Policy describing rollout cohorts, windows, and rollback criteria. Aliases:campaign policy, rollout cohorts, rollout windows
`ota campaign scheduled event is received` (EVT-OTA-CAMPAIGN-SCHEDULED)	Event raised when a fleet OTA campaign is scheduled. Aliases:campaign commands, campaign scheduled, ota commands, ota rollout commands
`ota status` (DATA-OTA-STATUS)	Device-level status for firmware download, apply, and rollback outcomes. Aliases:firmware apply, ota outcomes, rollback firmware apply, update outcomes, update/rollback status
`ota status reported event is received` (EVT-OTA-STATUS-REPORTED)	Event raised when device reports OTA apply status. Aliases:apply status, apply/rollback status, ota status reported, rollback status, status feedback, update status
`ota update agent` (FU-OTA-UPDATE-AGENT)	Downloads, validates, and applies signed firmware updates on machines. Aliases:device update agent
`platform operations` (FG-PLATFORM-OPERATIONS)	Capability area for runtime operations and device identity/secrets lifecycle. Aliases:fleet platform operations
`platform operator` (ACT-PLATFORM-OPERATOR)	Owns runtime deployment and operational controls. Aliases:platform ops

Term	Definition
`referenced element` (EM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
`requirement` (EM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
`runtime element` (EM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
`security analyst` (ACT-SECURITY-ANALYST)	Reviews identity, firmware integrity, and logging controls. Aliases:security reviewer
`signed firmware bundle` (DATA-FIRMWARE-BUNDLE)	Signed OTA firmware artifact distributed to machine cohorts for update apply. Aliases:firmware artifact, firmware bundle, firmware bundles, signed firmware bundles
`spoofed device identity` (AV-SPOOFED-DEVICE-IDENTITY)	Unauthorized actor impersonates a machine identity. Aliases:device spoofing
`state` (EM-STATE)	An authored lifecycle state used to model transition behavior, guards, and event-driven progression.
`telemetry ingested event is received` (EVT-TELEMETRY-INGESTED)	Event raised when cloud ingestion accepts telemetry. Aliases:fleet events, ingestion events, observability event fanout, observability events, telemetry ingested
`telemetry replay abuse` (AV-TELEMETRY-REPLAY-ABUSE)	Replayed telemetry floods ingestion and distorts reporting. Aliases:telemetry replay
`telemetry replay abuse is detected` (EVT-TELEMETRY-REPLAY-ABUSE-DETECTED)	Event raised when replayed telemetry behavior is detected and flagged as abuse. Aliases:replay abuse, telemetry replay detected
`trust boundary` (EM-TRUST-BOUNDARY)	An authored trust separation zone that marks policy, identity, or security control boundaries.
`update campaign orchestration` (FU-UPDATE-CAMPAIGN-ORCHESTRATION)	Selects target machine groups and coordinates OTA rollout phases. Aliases:campaign orchestration
`upward linking` (EM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
`verification check` (EM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.

4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



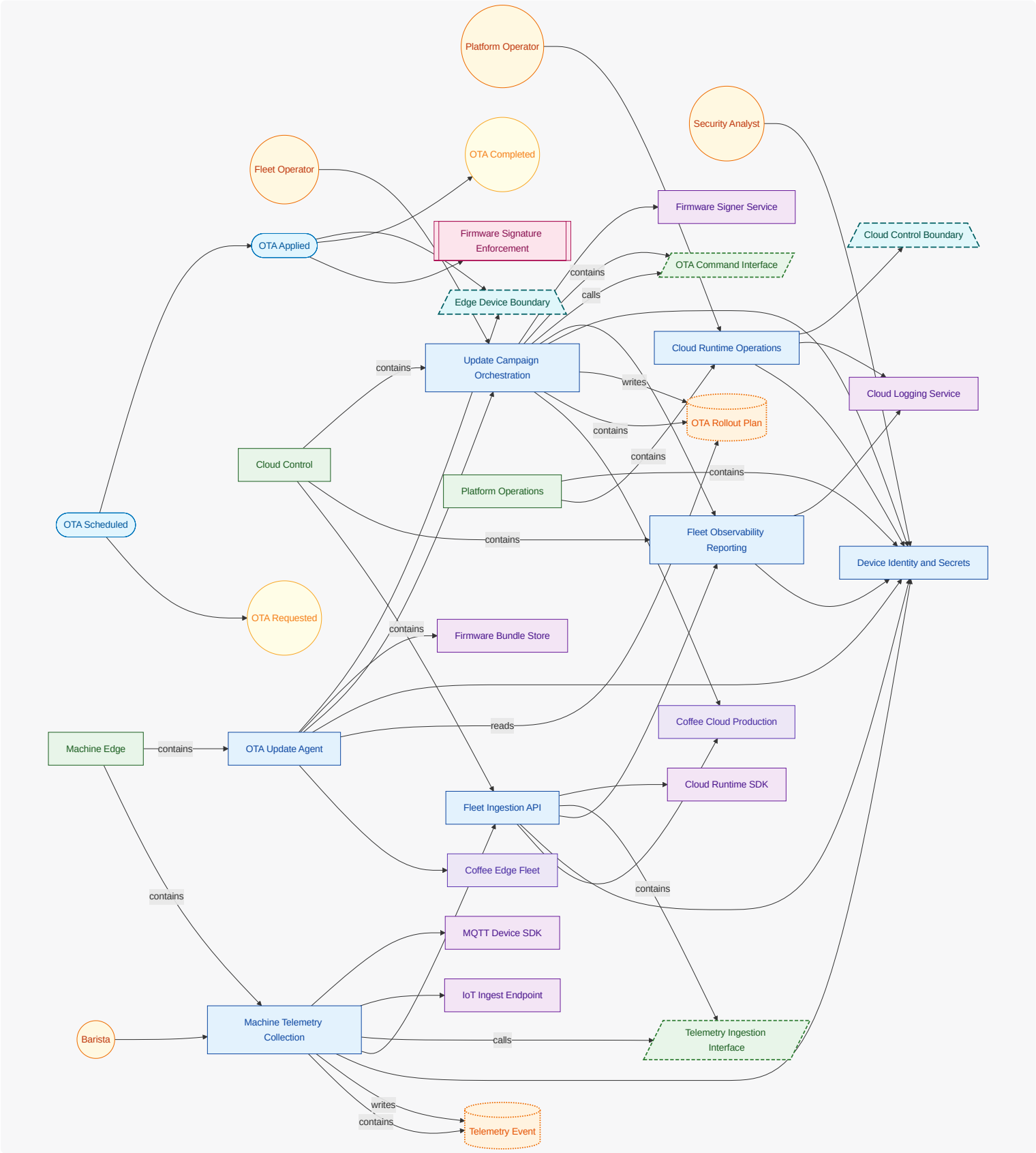
5. Architecture Intent View

Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

5.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

5.2. View-Scoped Projection



Node ID	Kind	Label
ACT-BARISTA	actor	Barista

Node ID	Kind	Label
ACT-FLEET-OPERATOR	actor	Fleet Operator
ACT-PLATFORM-OPERATOR	actor	Platform Operator
ACT-SECURITY-ANALYST	actor	Security Analyst
CTRL-COFFEE-FIRMWARE-SIGNATURE	control	Firmware Signature Enforcement
DO-COFFEE-OTA-PLAN	data_object	OTA Rollout Plan
DO-COFFEE-TELEMETRY-EVENT	data_object	Telemetry Event
DEP-COFFEE-CLOUD-PROD	deployment_target	Coffee Cloud Production
DEP-COFFEE-EDGE-FLEET	deployment_target	Coffee Edge Fleet
EVT-COFFEE-OTA-COMPLETED	event	OTA Completed
EVT-COFFEE-OTA-REQUESTED	event	OTA Requested
FG-CLOUD-CONTROL	functional_group	Cloud Control
FG-MACHINE-EDGE	functional_group	Machine Edge
FG-PLATFORM-OPERATIONS	functional_group	Platform Operations
FU-CLOUD-RUNTIME-OPERATIONS	functional_unit	Cloud Runtime Operations
FU-DEVICE-IDENTITY-SECRETS	functional_unit	Device Identity and Secrets
FU-FLEET-INGESTION-API	functional_unit	Fleet Ingestion API
FU-FLEET-OBSERVABILITY-REPORTING	functional_unit	Fleet Observability Reporting
FU-MACHINE-TELEMETRY-COLLECTION	functional_unit	Machine Telemetry Collection

Node ID	Kind	Label
`FU-OTA-UPDATE-AGENT`	functional_unit	`OTA Update Agent`
`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	functional_unit	`Update Campaign Orchestration`
`IF-COFFEE-OTA-COMMAND`	`interface`	`OTA Command Interface`
`IF-COFFEE-TELEMETRY-INGEST`	`interface`	`Telemetry Ingestion Interface`
`REF-CLOUD-LOGGING-SERVICE`	referenced_element	`Cloud Logging Service`
`REF-CLOUD-RUNTIME-SDK`	referenced_element	`Cloud Runtime SDK`
`REF-FIRMWARE-BUNDLE-STORE`	referenced_element	`Firmware Bundle Store`
`REF-FIRMWARE-SIGNER-SERVICE`	referenced_element	`Firmware Signer Service`
`REF-IOT-INGEST-ENDPOINT`	referenced_element	`IoT Ingest Endpoint`
`REF-MQTT-DEVICE-SDK`	referenced_element	`MQTT Device SDK`
`STATE-COFFEE-OTA-APPLIED`	`state`	`OTA Applied`
`STATE-COFFEE-OTA-SCHEDULED`	`state`	`OTA Scheduled`
`TB-COFFEE-CLOUD-CONTROL`	trust_boundary	`Cloud Control Boundary`
`TB-COFFEE-EDGE-DEVICE`	trust_boundary	`Edge Device Boundary`

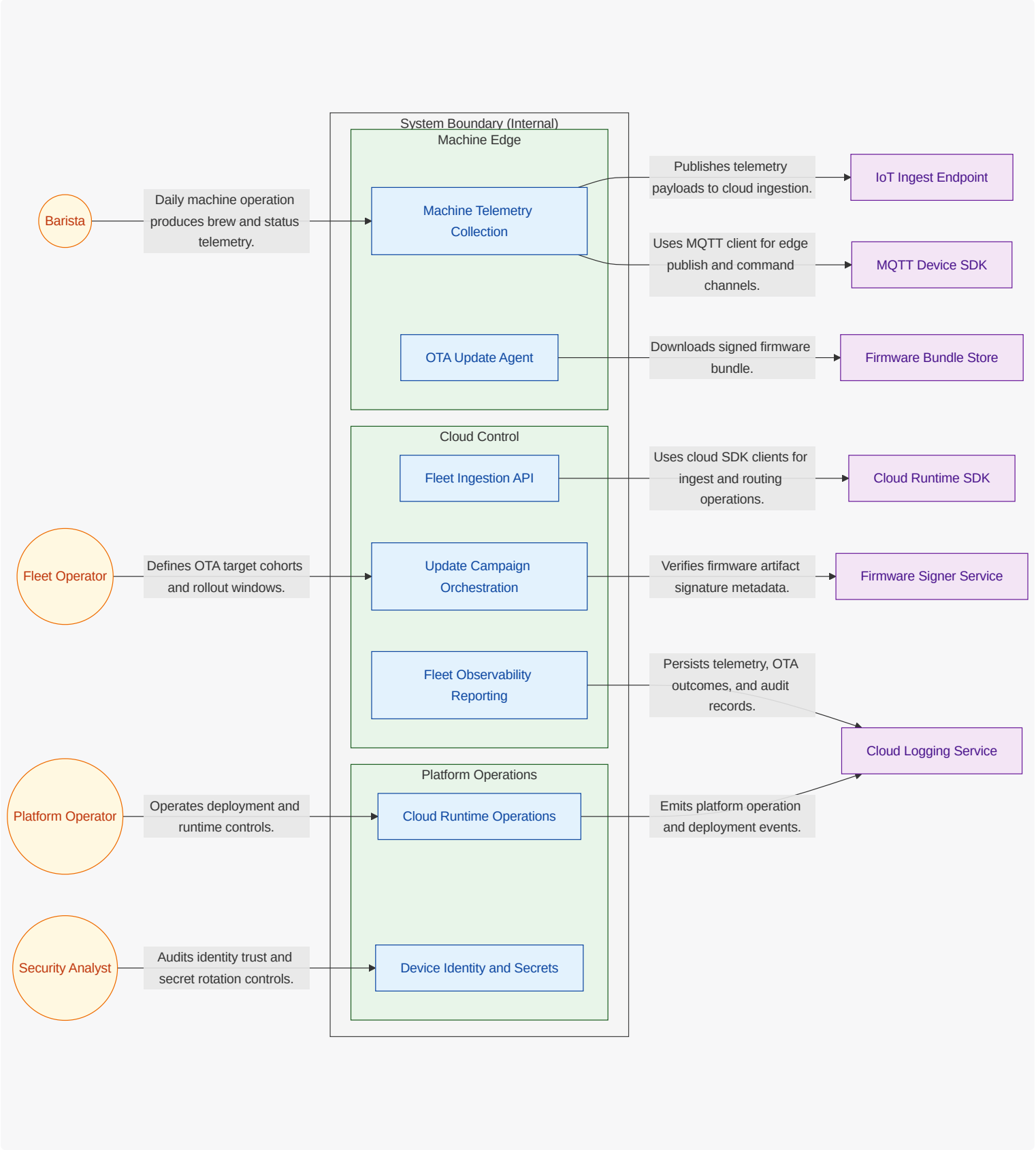
Mapping Type	From	To
bounded_by	`FU-CLOUD-RUNTIME-OPERATIONS`	`TB-COFFEE-CLOUD-CONTROL`
bounded_by	`FU-OTA-UPDATE-AGENT`	`TB-COFFEE-EDGE-DEVICE`
calls	`FU-MACHINE-TELEMETRY-COLLECTION`	`IF-COFFEE-TELEMETRY-INGEST`

Mapping Type	From	To
calls	FU-UPDATE-CAMPAIGN-ORCHESTRATION	IF-COFFEE-OTA-COMMAND
contains	FG-CLOUD-CONTROL	FU-FLEET-INGESTION-API
contains	FG-CLOUD-CONTROL	FU-FLEET-OBSERVABILITY-REPORTING
contains	FG-CLOUD-CONTROL	FU-UPDATE-CAMPAIGN-ORCHESTRATION
contains	FG-MACHINE-EDGE	FU-MACHINE-TELEMETRY-COLLECTION
contains	FG-MACHINE-EDGE	FU-OTA-UPDATE-AGENT
contains	FG-PLATFORM-OPERATIONS	FU-CLOUD-RUNTIME-OPERATIONS
contains	FG-PLATFORM-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS
contains	FU-FLEET-INGESTION-API	IF-COFFEE-TELEMETRY-INGEST
contains	FU-MACHINE-TELEMETRY-COLLECTION	DO-COFFEE-TELEMETRY-EVENT
contains	FU-UPDATE-CAMPAIGN-ORCHESTRATION	DO-COFFEE-OTA-PLAN
contains	FU-UPDATE-CAMPAIGN-ORCHESTRATION	IF-COFFEE-OTA-COMMAND
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-FLEET-INGESTION-API	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-FLEET-INGESTION-API	REF-CLOUD-RUNTIME-SDK
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-FLEET-INGESTION-API
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-IOT-INGEST-ENDPOINT
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-MQTT-DEVICE-SDK
depends_on	FU-OTA-UPDATE-AGENT	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
depends_on	FU-OTA-UPDATE-AGENT	REF-FIRMWARE-BUNDLE-STORE
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-DEVICE-IDENTITY-SECRETS

Mapping Type	From	To
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	REF-FIRMWARE-SIGNER-SERVICE
deployed_to	FU-FLEET-INGESTION-API	DEP-COFFEE-CLOUD-PROD
deployed_to	FU-OTA-UPDATE-AGENT	DEP-COFFEE-EDGE-FLEET
deployed_to	FU-UPDATE-CAMPAIGN-ORCHESTRATION	DEP-COFFEE-CLOUD-PROD
guarded_by	STATE-COFFEE-OTA-APPLIED	CTRL-COFFEE-FIRMWARE-SIGNATURE
guarded_by	STATE-COFFEE-OTA-APPLIED	TB-COFFEE-EDGE-DEVICE
interacts_with	ACT-BARISTA	FU-MACHINE-TELEMETRY-COLLECTION
interacts_with	ACT-FLEET-OPERATOR	FU-UPDATE-CAMPAIGN-ORCHESTRATION
interacts_with	ACT-PLATFORM-OPERATOR	FU-CLOUD-RUNTIME-OPERATIONS
interacts_with	ACT-SECURITY-ANALYST	FU-DEVICE-IDENTITY-SECRETS
reads	FU-OTA-UPDATE-AGENT	DO-COFFEE-OTA-PLAN
transitions_to	STATE-COFFEE-OTA-SCHEDULED	STATE-COFFEE-OTA-APPLIED
triggered_by	STATE-COFFEE-OTA-APPLIED	EVT-COFFEE-OTA-COMPLETED
triggered_by	STATE-COFFEE-OTA-SCHEDULED	EVT-COFFEE-OTA-REQUESTED
writes	FU-MACHINE-TELEMETRY-COLLECTION	DO-COFFEE-TELEMETRY-EVENT
writes	FU-UPDATE-CAMPAIGN-ORCHESTRATION	DO-COFFEE-OTA-PLAN

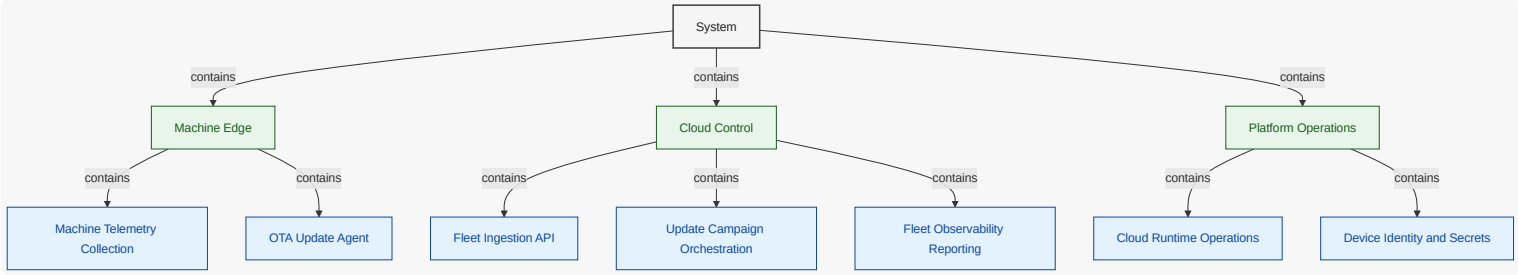
5.3. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.



5.4. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.


System
contains
contains
contains
Machine Edge
Cloud Control
Platform Operations
contains
contains
contains
contains
contains
contains
contains
Machine Telemetry Collection
OTA Update Agent
Fleet Ingestion API
Update Campaign Orchestration
Fleet Observability Reporting
Cloud Runtime Operations
Device Identity and Secrets

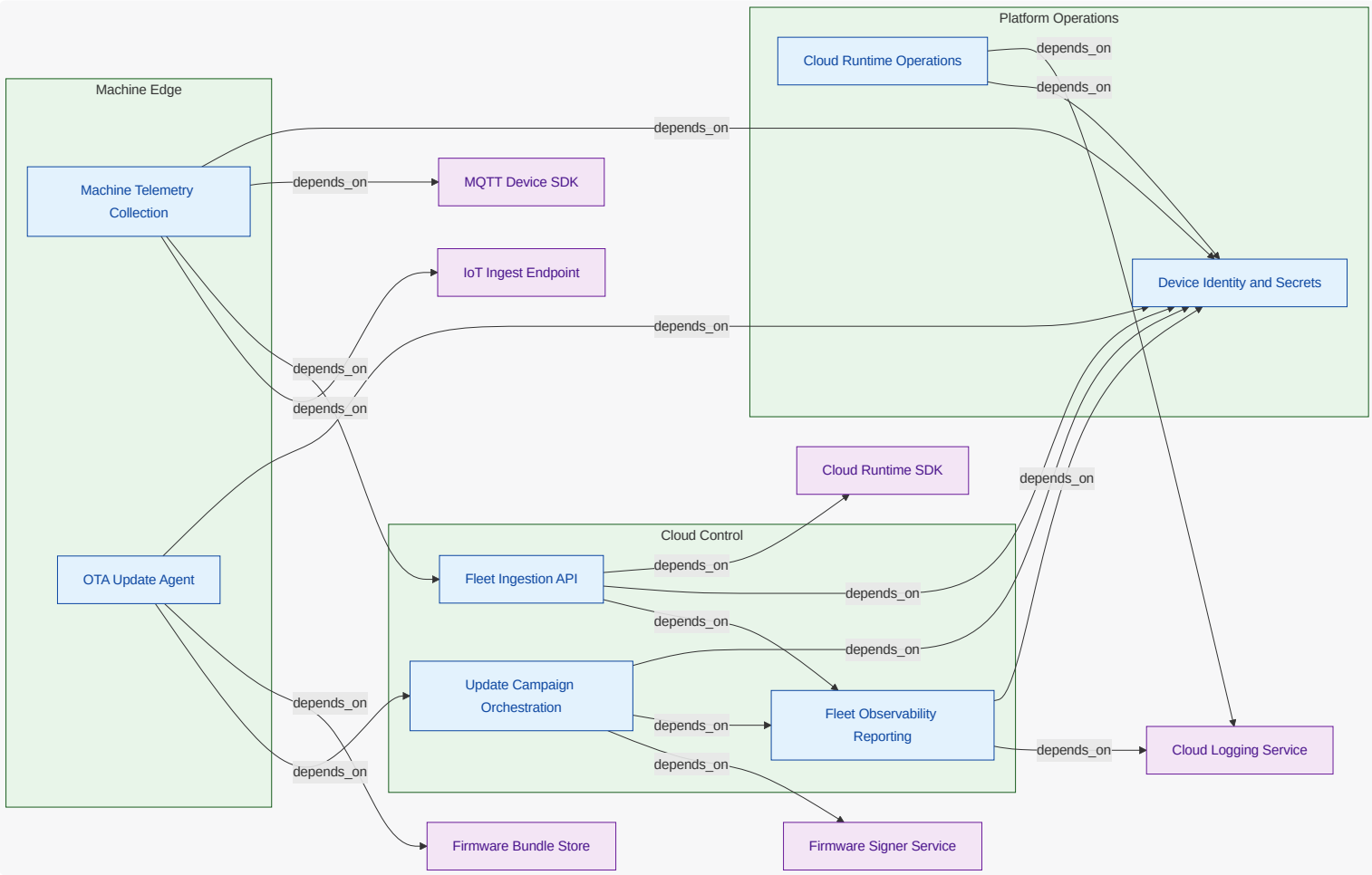
5.5. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

	Fleet Ingestion API	
Machine Telemetry Collection	Fleet Observability Reporting	Cloud Runtime Operations
OTA Update Agent	Update Campaign Orchestration	Device Identity and Secrets
Machine Edge	Cloud Control	Platform Operations

5.6. Functional Unit Dependency Diagram

What this diagram shows: dependency paths between functional units (grouped in Functional Group containers) and key referenced elements. Arrow meaning: `A -→|depends_on| B` means unit `A` requires behavior or data from `B` to fulfill its responsibility.



5.7. Functional Groups and Units (View Scoped)

5.7.1. Machine Edge (FG-MACHINE-EDGE)

Device-local `telemetry` and OTA execution responsibilities.

`Machine Edge` owns behavior that executes directly on coffee machines. It gathers `telemetry`, applies OTA updates, and reports local status to cloud boundaries.

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Collects brew/device `telemetry` and transmits cloud-bound `telemetry payloads`.

Owned Responsibility

functional responsibility for `machine telemetry collection`; includes decision and orchestration `flow` to `Device Identity and Secrets` and decision and orchestration `flow` to `Fleet Ingestion API`

Key Collaborators

`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

Linked Requirements

`REQ-COF-001`, `REQ-COF-006`, `REQ-COF-011`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

Downloads `signed firmware bundles`, validates integrity, applies updates, and reports status.

Owned Responsibility

functional responsibility for `ota update agent`; includes decision and orchestration `flow` to `Device Identity and Secrets` and decision and orchestration `flow` to `Firmware Bundle Store`

Key Collaborators

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

Linked Requirements

`REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-010`

5.7.2. Cloud Control (FG-CLOUD-CONTROL)

Central cloud ingestion, campaign `control`, and fleet reporting.

`Cloud Control` owns central APIs and coordination logic for fleet `telemetry` and `OTA rollouts`. It provides global fleet visibility and deterministic rollout decisions.

Fleet Ingestion API (FU-FLEET-INGESTION-API)

Receives `telemetry` and `machine status`, applies validation, and emits `ingestion events`.

Owned Responsibility

functional responsibility for `fleet ingestion api`; includes decision and orchestration `flow` to `Cloud Runtime SDK` and decision and orchestration `flow` to `Device Identity and Secrets`

Key Collaborators

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Linked Requirements

`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-011`, `REQ-COF-012`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Aggregates `telemetry` and `OTA outcomes` into dashboards, alerts, and `audit records`.

Owned Responsibility

functional responsibility for `fleet observability reporting`; includes decision and orchestration `flow` to `Cloud Logging Service` and decision and orchestration `flow` to `Device Identity and Secrets`

Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

Linked Requirements

`REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Plans OTA `rollout cohorts`, drives update commands, and handles rollback `flow` decisions.

Owned Responsibility

functional responsibility for `update campaign orchestration`; includes decision and orchestration `flow` to `Device Identity and Secrets` and decision and orchestration `flow` to `Firmware Signer Service`

Key Collaborators

`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

Linked Requirements

`REQ-COF-003`, `REQ-COF-005`, `REQ-COF-010`

5.7.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Cloud runtime operations` plus identity and secret-management controls.

`Platform Operations` owns deployability, runtime guardrails, and secure identity/config lifecycle. It ensures cloud and device `trust boundaries` remain enforceable over time.

Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

Owned Responsibility

functional responsibility for `cloud runtime operations`; includes decision and orchestration `flow` to `Cloud Logging Service` and decision and orchestration `flow` to `Device Identity and Secrets`

Key Collaborators

`Cloud Logging Service`, `Device Identity and Secrets`

Linked Requirements

`REQ-COF-008`, `REQ-COF-012`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

Owned Responsibility

functional responsibility for `device identity and secrets`

Key Collaborators

none

Linked Requirements

`REQ-COF-004`, `REQ-COF-007`

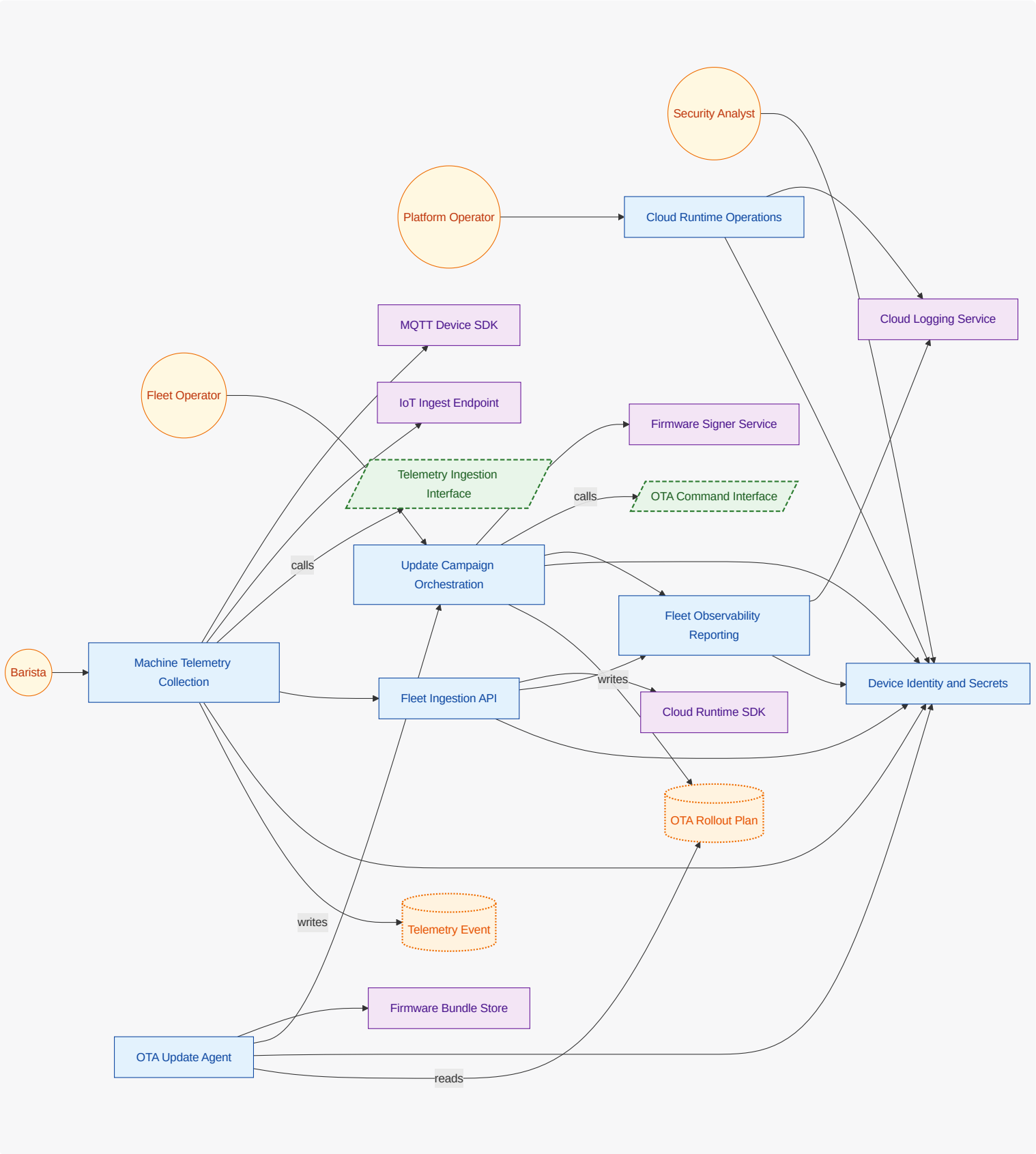
6. Communication View

Show who communicates with whom across key request paths, including `interface` edges and dependency handoffs.

6.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request `flow`?

6.2. View-Scoped Projection



Node ID	Kind	Label
ACT-BARISTA	actor	Barista

Node ID	Kind	Label
ACT-FLEET-OPERATOR	actor	Fleet Operator
ACT-PLATFORM-OPERATOR	actor	Platform Operator
ACT-SECURITY-ANALYST	actor	Security Analyst
DO-COFFEE-OTA-PLAN	data_object	OTA Rollout Plan
DO-COFFEE-TELEMETRY-EVENT	data_object	Telemetry Event
FU-CLOUD-RUNTIME-OPERATIONS	functional_unit	Cloud Runtime Operations
FU-DEVICE-IDENTITY-SECRETS	functional_unit	Device Identity and Secrets
FU-FLEET-INGESTION-API	functional_unit	Fleet Ingestion API
FU-FLEET-OBSERVABILITY-REPORTING	functional_unit	Fleet Observability Reporting
FU-MACHINE-TELEMETRY-COLLECTION	functional_unit	Machine Telemetry Collection
FU-OTA-UPDATE-AGENT	functional_unit	OTA Update Agent
FU-UPDATE-CAMPAIGN-ORCHESTRATION	functional_unit	Update Campaign Orchestration
IF-COFFEE-OTA-COMMAND	interface	OTA Command Interface
IF-COFFEE-TELEMETRY-INGEST	interface	Telemetry Ingestion Interface
REF-CLOUD-LOGGING-SERVICE	referenced_element	Cloud Logging Service
REF-CLOUD-RUNTIME-SDK	referenced_element	Cloud Runtime SDK
REF-FIRMWARE-BUNDLE-STORE	referenced_element	Firmware Bundle Store

Node ID	Kind	Label
REF-FIRMWARE-SIGNER-SERVICE	referenced_element	Firmware Signer Service
REF-IOT-INGEST-ENDPOINT	referenced_element	IoT Ingest Endpoint
REF-MQTT-DEVICE-SDK	referenced_element	MQTT Device SDK

Mapping Type	From	To
calls	FU-MACHINE-TELEMETRY-COLLECTION	IF-COFFEE-TELEMETRY-INGEST
calls	FU-UPDATE-CAMPAIGN-ORCHESTRATION	IF-COFFEE-OTA-COMMAND
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-FLEET-INGESTION-API	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-FLEET-INGESTION-API	REF-CLOUD-RUNTIME-SDK
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-FLEET-INGESTION-API
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-IOT-INGEST-ENDPOINT
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-MQTT-DEVICE-SDK
depends_on	FU-OTA-UPDATE-AGENT	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
depends_on	FU-OTA-UPDATE-AGENT	REF-FIRMWARE-BUNDLE-STORE
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	REF-FIRMWARE-SIGNER-SERVICE
interacts_with	ACT-BARISTA	FU-MACHINE-TELEMETRY-COLLECTION
interacts_with	ACT-FLEET-OPERATOR	FU-UPDATE-CAMPAIGN-ORCHESTRATION

Mapping Type	From	To
interacts_with	`ACT-PLATFORM-OPERATOR`	`FU-CLOUD-RUNTIME-OPERATIONS`
interacts_with	`ACT-SECURITY-ANALYST`	`FU-DEVICE-IDENTITY-SECRETS`
reads	`FU-OTA-UPDATE-AGENT`	`DO-COFFEE-OTA-PLAN`
writes	`FU-MACHINE-TELEMETRY-COLLECTION`	`DO-COFFEE-TELEMETRY-EVENT`
writes	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`DO-COFFEE-OTA-PLAN`

6.3. Coverage Gaps

Coverage summary: 4/7 units have direct evidence coverage in this view.

- `Device Identity and Secrets` has no direct inferred runtime/deployment evidence yet.
- `Machine Telemetry Collection` has no direct inferred runtime/deployment evidence yet.
- `OTA Update Agent` has no direct inferred runtime/deployment evidence yet.

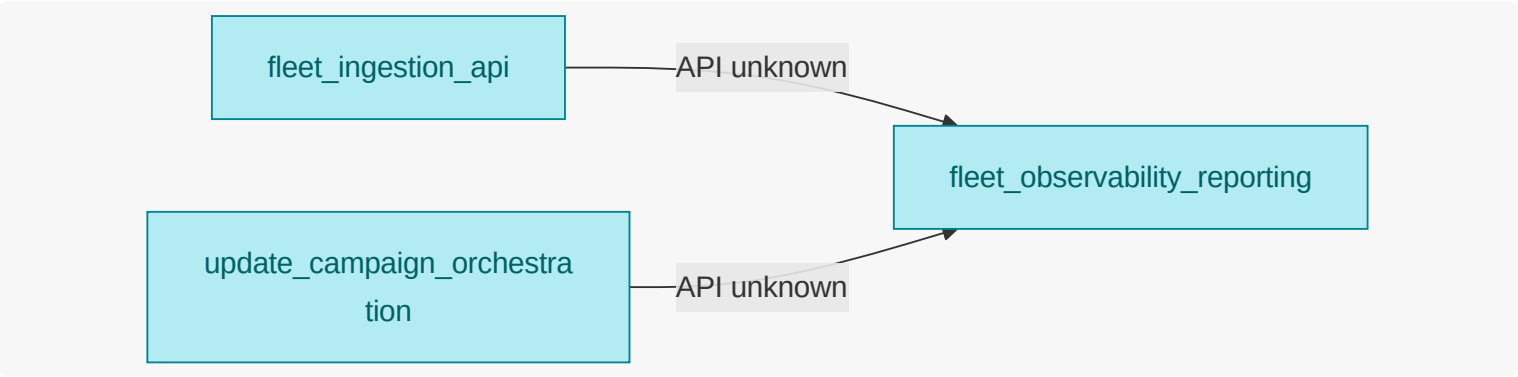
6.4. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/`event`/callback) to `authored mappings` where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

6.5. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE** This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.



6.6. Functional Groups and Units (View Scoped)

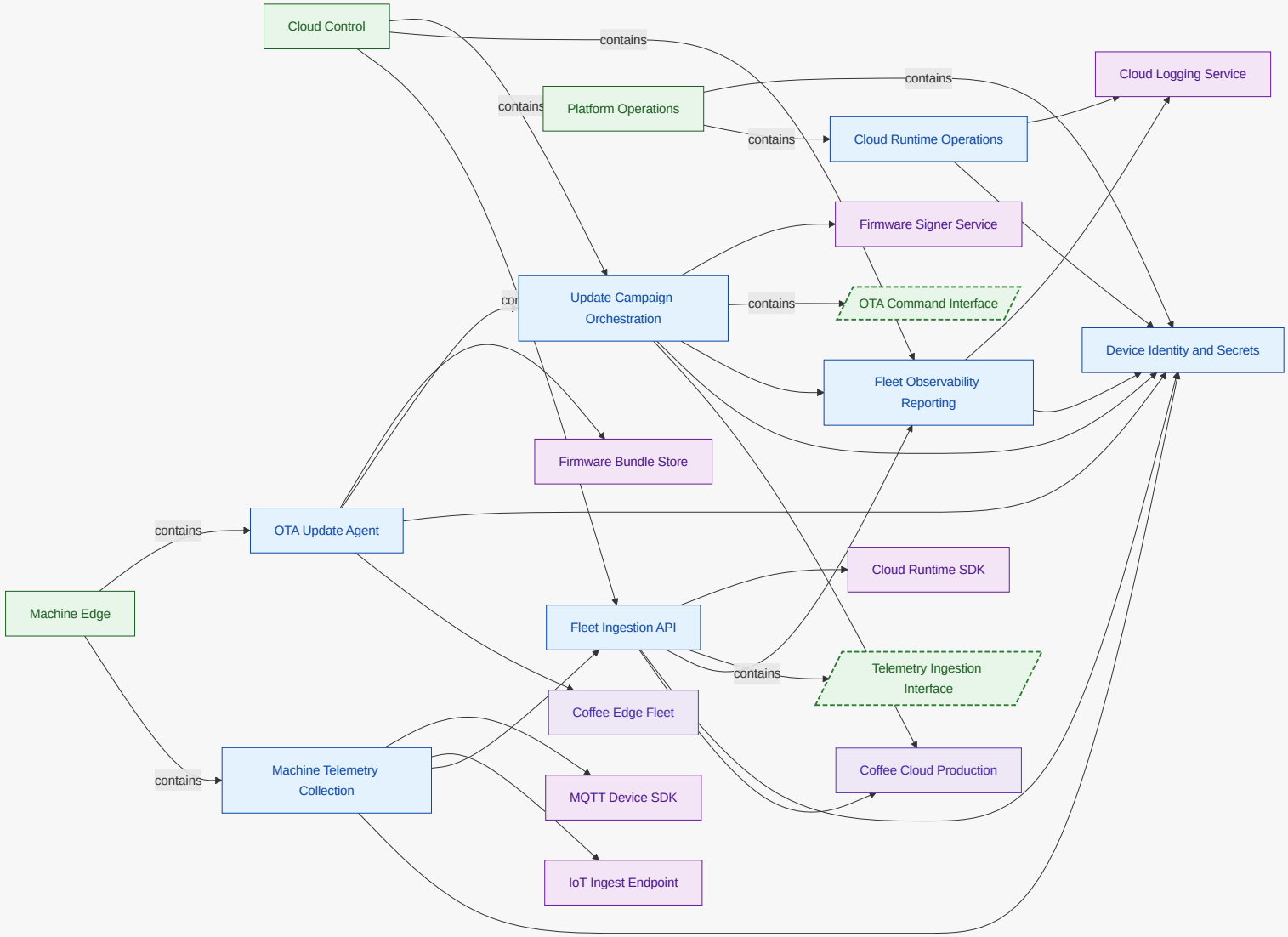
7. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and `platform operations`.

7.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

7.2. View-Scoped Projection



Node ID	Kind	Label
DEP-COFFEE-CLOUD-PROD	deployment_target	Coffee Cloud Production
DEP-COFFEE-EDGE-FLEET	deployment_target	Coffee Edge Fleet
FG-CLOUD-CONTROL	functional_group	Cloud Control
FG-MACHINE-EDGE	functional_group	Machine Edge
FG-PLATFORM-OPERATIONS	functional_group	Platform Operations
FU-CLOUD-RUNTIME-OPERATIONS	functional_unit	Cloud Runtime Operations
FU-DEVICE-IDENTITY-SECRETS	functional_unit	Device Identity and Secrets
FU-FLEET-INGESTION-API	functional_unit	Fleet Ingestion API
FU-FLEET-OBSERVABILITY-REPORTING	functional_unit	Fleet Observability Reporting
FU-MACHINE-TELEMETRY-COLLECTION	functional_unit	Machine Telemetry Collection
FU-OTA-UPDATE-AGENT	functional_unit	OTA Update Agent
FU-UPDATE-CAMPAIGN-ORCHESTRATION	functional_unit	Update Campaign Orchestration
IF-COFFEE-OTA-COMMAND	interface	OTA Command Interface
IF-COFFEE-TELEMETRY-INGEST	interface	Telemetry Ingestion Interface
REF-CLOUD-LOGGING-SERVICE	referenced_element	Cloud Logging Service
REF-CLOUD-RUNTIME-SDK	referenced_element	Cloud Runtime SDK
REF-FIRMWARE-BUNDLE-STORE	referenced_element	Firmware Bundle Store

Node ID	Kind	Label
REF-FIRMWARE-SIGNER-SERVICE	referenced_element	Firmware Signer Service
REF-IOT-INGEST-ENDPOINT	referenced_element	IoT Ingest Endpoint
REF-MQTT-DEVICE-SDK	referenced_element	MQTT Device SDK

Mapping Type	From	To
contains	FG-CLOUD-CONTROL	FU-FLEET-INGESTION-API
contains	FG-CLOUD-CONTROL	FU-FLEET-OBSERVABILITY-REPORTING
contains	FG-CLOUD-CONTROL	FU-UPDATE-CAMPAIGN-ORCHESTRATION
contains	FG-MACHINE-EDGE	FU-MACHINE-TELEMETRY-COLLECTION
contains	FG-MACHINE-EDGE	FU-OTA-UPDATE-AGENT
contains	FG-PLATFORM-OPERATIONS	FU-CLOUD-RUNTIME-OPERATIONS
contains	FG-PLATFORM-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS
contains	FU-FLEET-INGESTION-API	IF-COFFEE-TELEMETRY-INGEST
contains	FU-UPDATE-CAMPAIGN-ORCHESTRATION	IF-COFFEE-OTA-COMMAND
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-FLEET-INGESTION-API	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-FLEET-INGESTION-API	REF-CLOUD-RUNTIME-SDK
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-FLEET-INGESTION-API
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-IOT-INGEST-ENDPOINT
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-MQTT-DEVICE-SDK
depends_on	FU-OTA-UPDATE-AGENT	FU-DEVICE-IDENTITY-SECRETS

Mapping Type	From	To
depends_on	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
depends_on	FU-OTA-UPDATE-AGENT	REF-FIRMWARE-BUNDLE-STORE
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	REF-FIRMWARE-SIGNER-SERVICE
deployed_to	FU-FLEET-INGESTION-API	DEP-COFFEE-CLOUD-PROD
deployed_to	FU-OTA-UPDATE-AGENT	DEP-COFFEE-EDGE-FLEET
deployed_to	FU-UPDATE-CAMPAIGN-ORCHESTRATION	DEP-COFFEE-CLOUD-PROD

7.3. Coverage Gaps

Coverage summary: 4/7 units have direct evidence coverage in this view.

- `Device Identity and Secrets` has no direct inferred runtime/deployment evidence yet.
- `Machine Telemetry Collection` has no direct inferred runtime/deployment evidence yet.
- `OTA Update Agent` has no direct inferred runtime/deployment evidence yet.

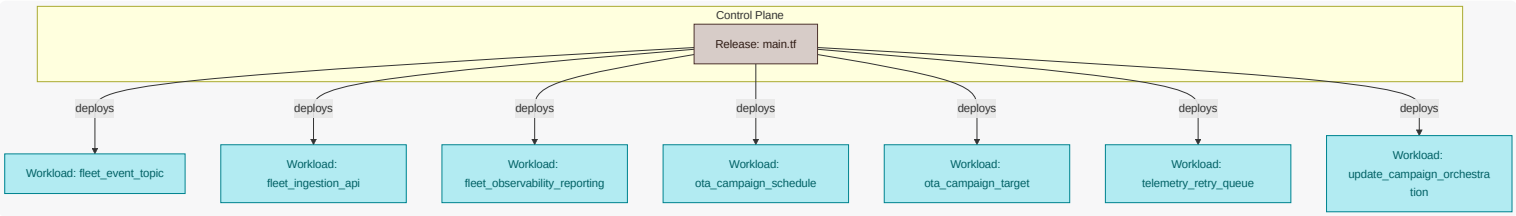
7.4. Recommended Next Evidence Additions

- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment `control` artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

7.5. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

> **NOTE**
> This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.

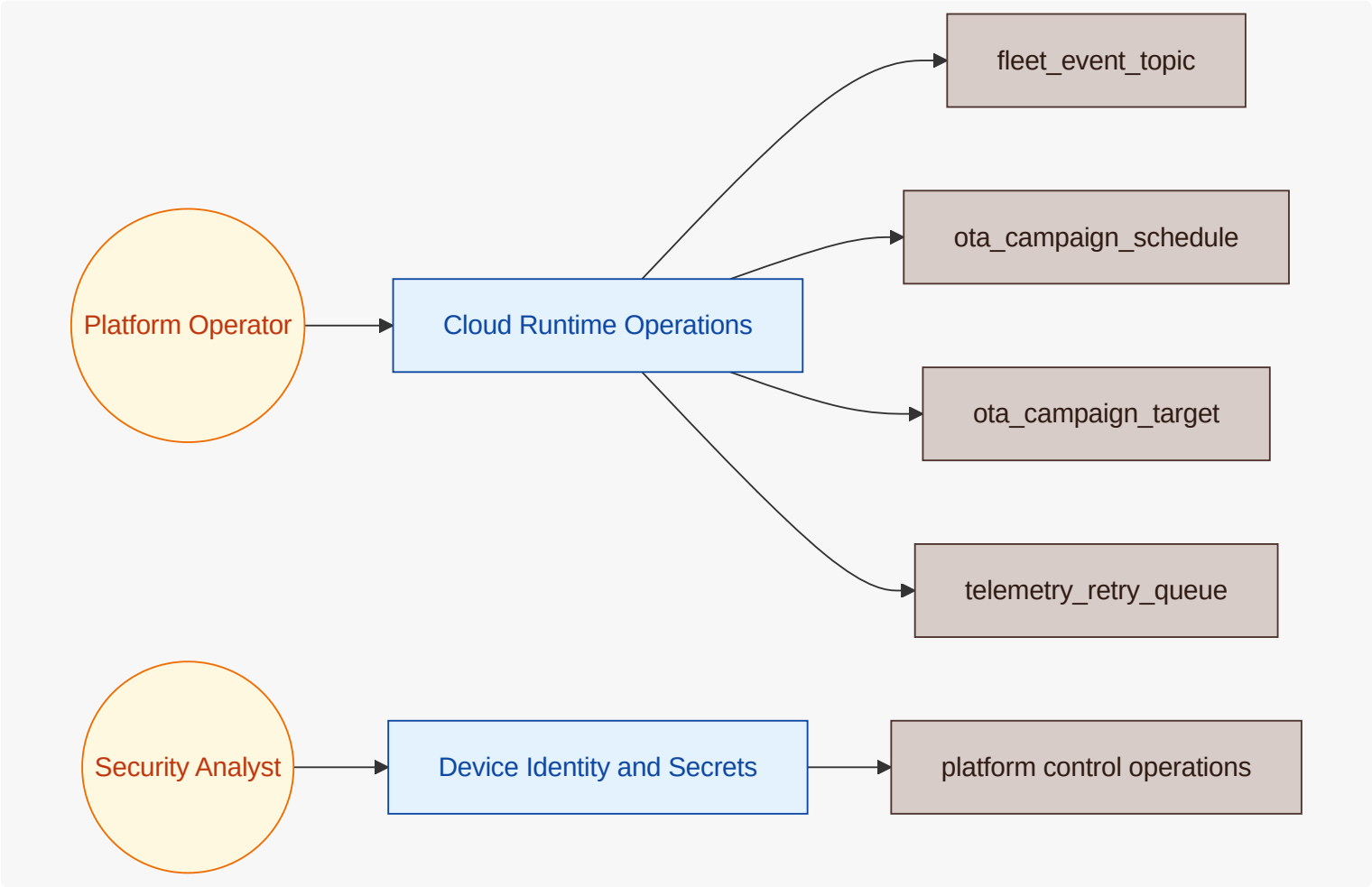


7.6. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

NOTE

Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.



7.7. Functional Groups and Units (View Scoped)

7.7.1. Machine Edge (FG-MACHINE-EDGE)

Machine-edge software artifacts are rolled independently from `cloud control` to isolate device risk.

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Machine Telemetry Collection Deployment

Deployed as edge software component with independent release cadence.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-001`, `REQ-COF-006`, `REQ-COF-011`
Deployment dependencies	`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`
Inferred deployment evidence	code modules: `machine_telemetry`

OTA Update Agent (FU-OTA-UPDATE-AGENT)

OTA Update Agent Deployment

Versioned and rolled independently from `telemetry` pipeline components.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-010`
Deployment dependencies	`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`
Inferred deployment evidence	code modules: `ota_update_agent`

7.7.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud-control units deploy independently to reduce blast radius during campaign logic changes.

Fleet Ingestion API (FU-FLEET-INGESTION-API)

Fleet Ingestion API Deployment

Deployed as independent cloud ingestion boundary to isolate scaling concerns.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-011`, `REQ-COF-012`
Deployment dependencies	`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`
Inferred deployment evidence	runtime: `fleet_ingestion_api`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Fleet Observability Reporting Deployment

Deployed as reporting plane with independent scaling and retention controls.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-002`, `REQ-COF-005`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`
Deployment dependencies	`Cloud Logging Service`, `Device Identity and Secrets`
Inferred deployment evidence	runtime: `fleet_observability_reporting`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Update Campaign Orchestration Deployment

Isolated deployment supports safe OTA policy evolution.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-003`, `REQ-COF-005`, `REQ-COF-010`
Deployment dependencies	`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`
Inferred deployment evidence	runtime: `update_campaign_orchestration`

7.7.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform operations` governs release sequencing, runtime placement, and operational guardrails.

Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Cloud Runtime Operations Deployment

Owns source-to-runtime release controls and rollback guardrails.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-008`, `REQ-COF-012`
Deployment dependencies	`Cloud Logging Service`, `Device Identity and Secrets`
Inferred deployment evidence	runtime: `fleet_event_topic`, `ota_campaign_schedule`, `ota_campaign_target`, `telemetry_retry_queue`

Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Device Identity and Secrets Deployment

Managed as shared control-plane boundary with controlled key rotation.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-COF-004`, `REQ-COF-007`
Deployment dependencies	none
Inferred deployment evidence	code modules: `device_identity_secrets`

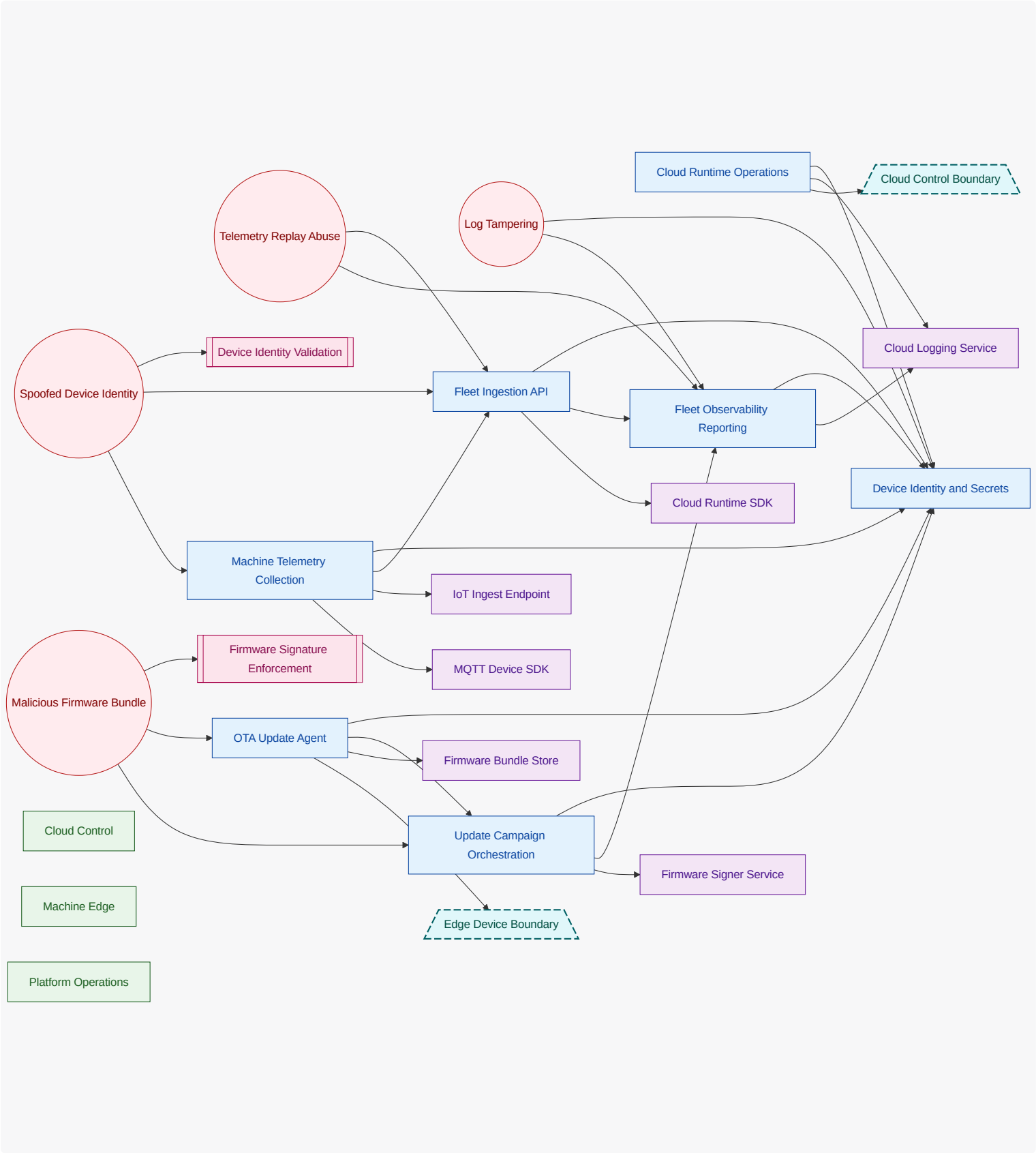
8. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence.

8.1. What This View Answers

- What are the main `trust boundaries` and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

8.2. View-Scoped Projection



Node ID	Kind	Label
AV-LOG-TAMPERING	attack_vector	Log Tampering

Node ID	Kind	Label
AV-MALICIOUS-FIRMWARE-BUNDLE	attack_vector	Malicious Firmware Bundle
AV-SPOOFED-DEVICE-IDENTITY	attack_vector	Spoofed Device Identity
AV-TELEMETRY-REPLAY-ABUSE	attack_vector	Telemetry Replay Abuse
CTRL-COFFEE-DEVICE-IDENTITY	control	Device Identity Validation
CTRL-COFFEE-FIRMWARE-SIGNATURE	control	Firmware Signature Enforcement
FG-CLOUD-CONTROL	functional_group	Cloud Control
FG-MACHINE-EDGE	functional_group	Machine Edge
FG-PLATFORM-OPERATIONS	functional_group	Platform Operations
FU-CLOUD-RUNTIME-OPERATIONS	functional_unit	Cloud Runtime Operations
FU-DEVICE-IDENTITY-SECRETS	functional_unit	Device Identity and Secrets
FU-FLEET-INGESTION-API	functional_unit	Fleet Ingestion API
FU-FLEET-OBSERVABILITY-REPORTING	functional_unit	Fleet Observability Reporting
FU-MACHINE-TELEMETRY-COLLECTION	functional_unit	Machine Telemetry Collection
FU-OTA-UPDATE-AGENT	functional_unit	OTA Update Agent
FU-UPDATE-CAMPAIGN-ORCHESTRATION	functional_unit	Update Campaign Orchestration
REF-CLOUD-LOGGING-SERVICE	referenced_element	Cloud Logging Service
REF-CLOUD-RUNTIME-SDK	referenced_element	Cloud Runtime SDK

Node ID	Kind	Label
REF-FIRMWARE-BUNDLE-STORE	referenced_element	Firmware Bundle Store
REF-FIRMWARE-SIGNER-SERVICE	referenced_element	Firmware Signer Service
REF-IOT-INGEST-ENDPOINT	referenced_element	IoT Ingest Endpoint
REF-MQTT-DEVICE-SDK	referenced_element	MQTT Device SDK
TB-COFFEE-CLOUD-CONTROL	trust_boundary	Cloud Control Boundary
TB-COFFEE-EDGE-DEVICE	trust_boundary	Edge Device Boundary

Mapping Type	From	To
bounded_by	FU-CLOUD-RUNTIME-OPERATIONS	TB-COFFEE-CLOUD-CONTROL
bounded_by	FU-OTA-UPDATE-AGENT	TB-COFFEE-EDGE-DEVICE
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-CLOUD-RUNTIME-OPERATIONS	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-FLEET-INGESTION-API	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING
depends_on	FU-FLEET-INGESTION-API	REF-CLOUD-RUNTIME-SDK
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-FLEET-OBSERVABILITY-REPORTING	REF-CLOUD-LOGGING-SERVICE
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	FU-FLEET-INGESTION-API
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-IOT-INGEST-ENDPOINT
depends_on	FU-MACHINE-TELEMETRY-COLLECTION	REF-MQTT-DEVICE-SDK
depends_on	FU-OTA-UPDATE-AGENT	FU-DEVICE-IDENTITY-SECRETS
depends_on	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
depends_on	FU-OTA-UPDATE-AGENT	REF-FIRMWARE-BUNDLE-STORE
depends_on	FU-UPDATE-CAMPAIGN-ORCHESTRATION	FU-DEVICE-IDENTITY-SECRETS

Mapping Type	From	To
depends_on	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`FU-FLEET-OBSERVABILITY-REPORTING`
depends_on	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`REF-FIRMWARE-SIGNER-SERVICE`
mitigated_by	`AV-MALICIOUS-FIRMWARE-BUNDLE`	`CTRL-COFFEE-FIRMWARE-SIGNATURE`
mitigated_by	`AV-SPOOFED-DEVICE-IDENTITY`	`CTRL-COFFEE-DEVICE-IDENTITY`
targets	`AV-LOG-TAMPERING`	`FU-DEVICE-IDENTITY-SECRETS`
targets	`AV-LOG-TAMPERING`	`FU-FLEET-OBSERVABILITY-REPORTING`
targets	`AV-MALICIOUS-FIRMWARE-BUNDLE`	`FU-OTA-UPDATE-AGENT`
targets	`AV-MALICIOUS-FIRMWARE-BUNDLE`	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`
targets	`AV-SPOOFED-DEVICE-IDENTITY`	`FU-FLEET-INGESTION-API`
targets	`AV-SPOOFED-DEVICE-IDENTITY`	`FU-MACHINE-TELEMETRY-COLLECTION`
targets	`AV-TELEMETRY-REPLAY-ABUSE`	`FU-FLEET-INGESTION-API`
targets	`AV-TELEMETRY-REPLAY-ABUSE`	`FU-FLEET-OBSERVABILITY-REPORTING`

8.3. Coverage Gaps

Coverage summary: 6/7 units have direct evidence coverage in this view.

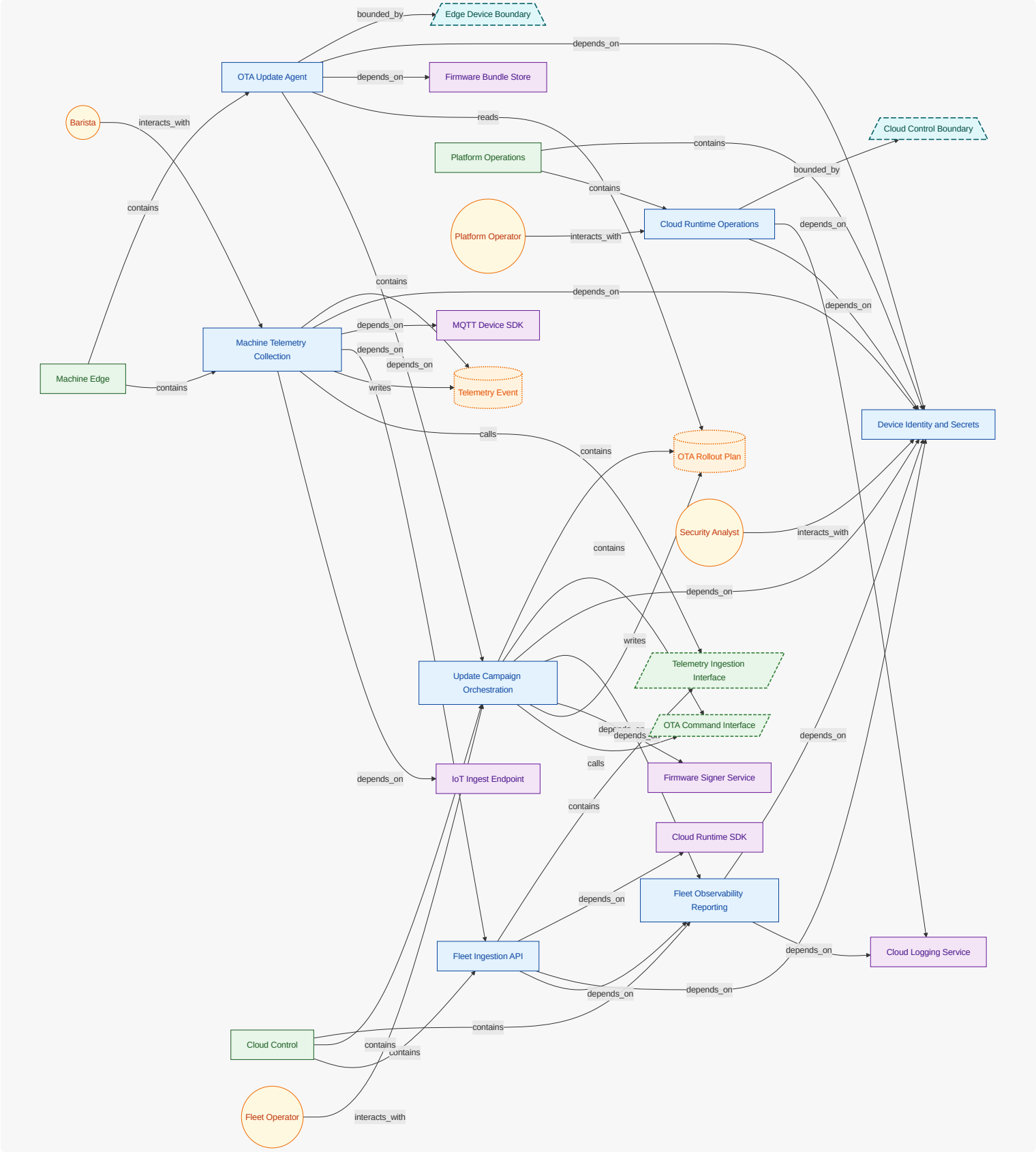
- `Cloud Runtime Operations` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

8.4. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/`events`) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

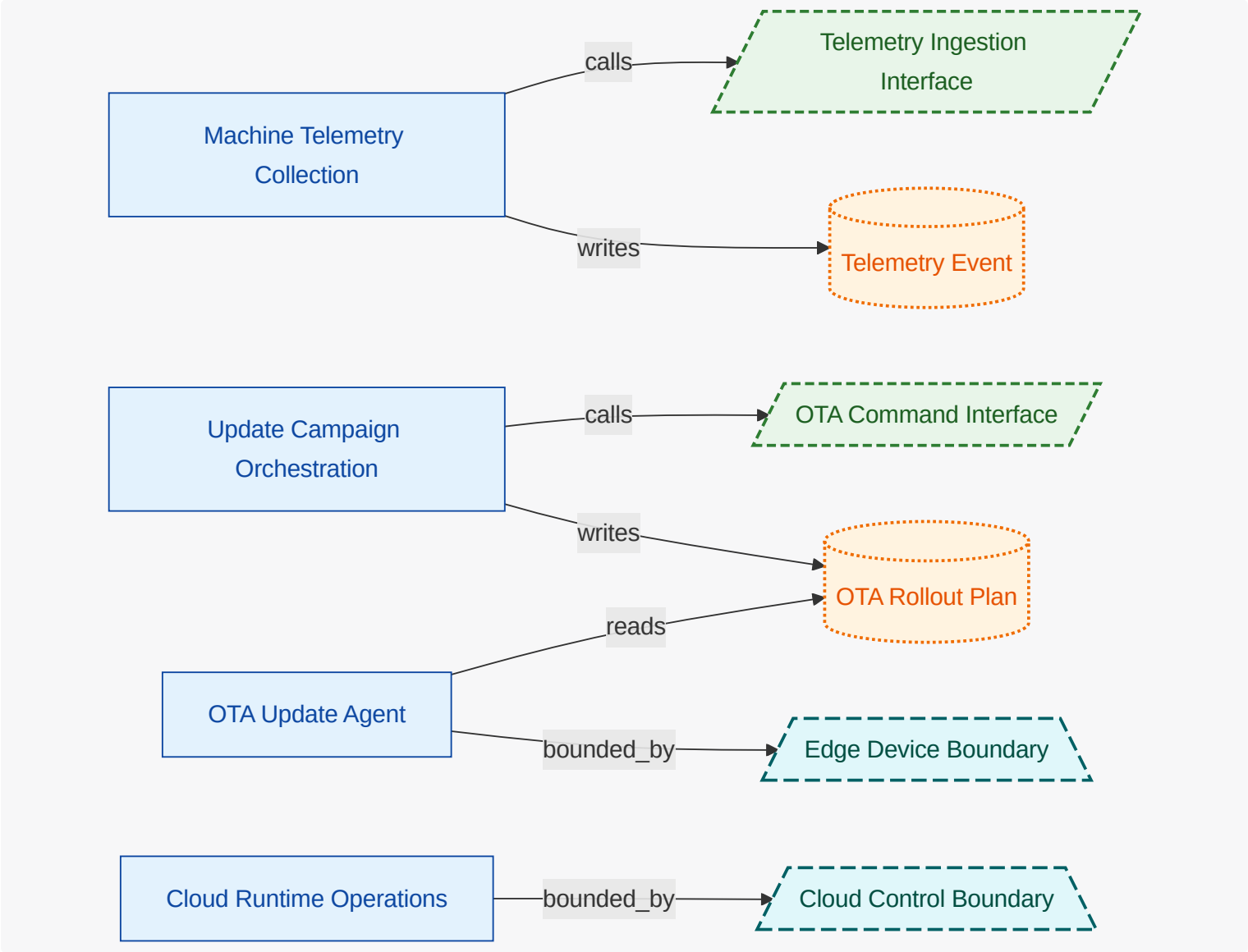
8.5. Threat Model Context Diagram

What this diagram shows: OWASP-style context view of actors, core processes, interfaces, and trust boundary-relevant entities.



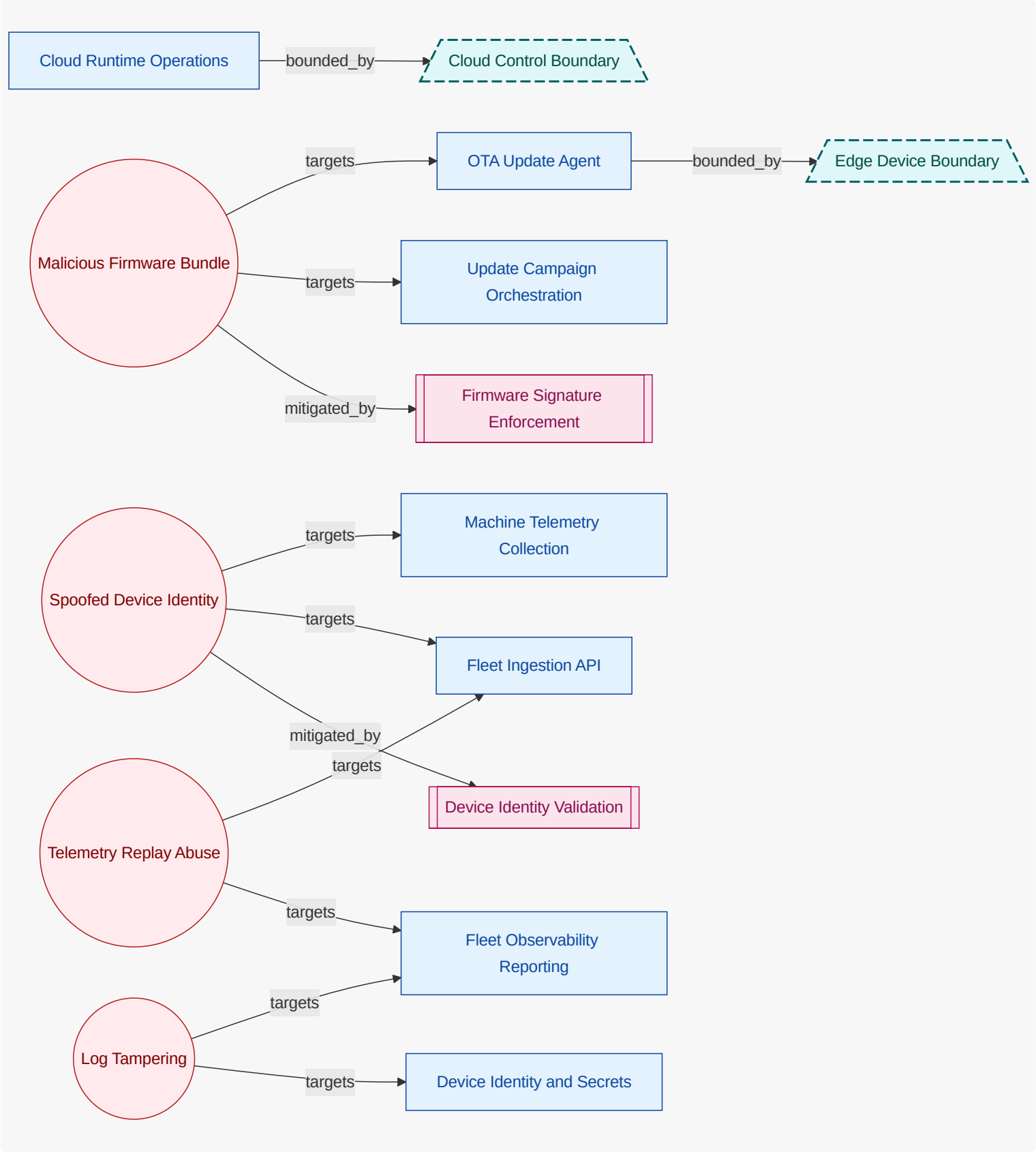
8.6. Threat Model Data Flow Diagram

What this diagram shows: primary security-relevant data flows (reads/writes/publishes/subscribes/streams) between processes and data objects.



8.7. Threat-to-Control Overlay Diagram

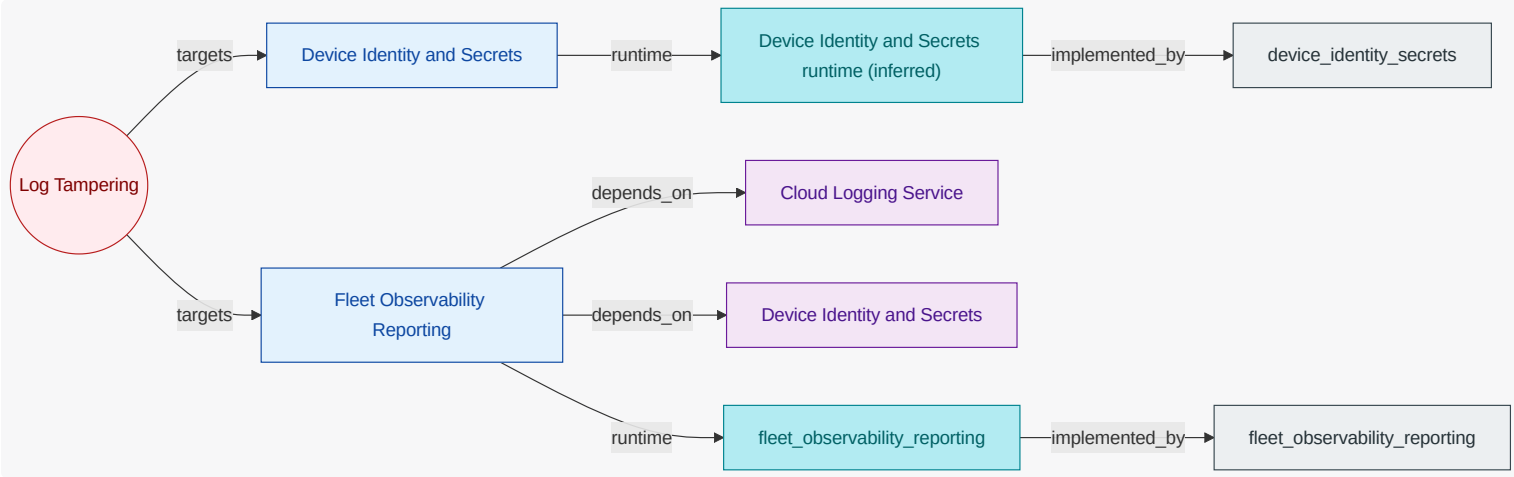
What this diagram shows: attack vectors, targeted components, and mitigation/boundary relationships.



8.8. Attack Vector Chapters

8.8.1. Log Tampering (AV-LOG-TAMPERING)

Attempts to suppress or alter audit/operational logs.



Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Protects credential lifecycle and enforces least-privilege access patterns.

Attack Surface

none

Security Evidence Present

code modules: `device_identity_secrets`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

Attack Surface

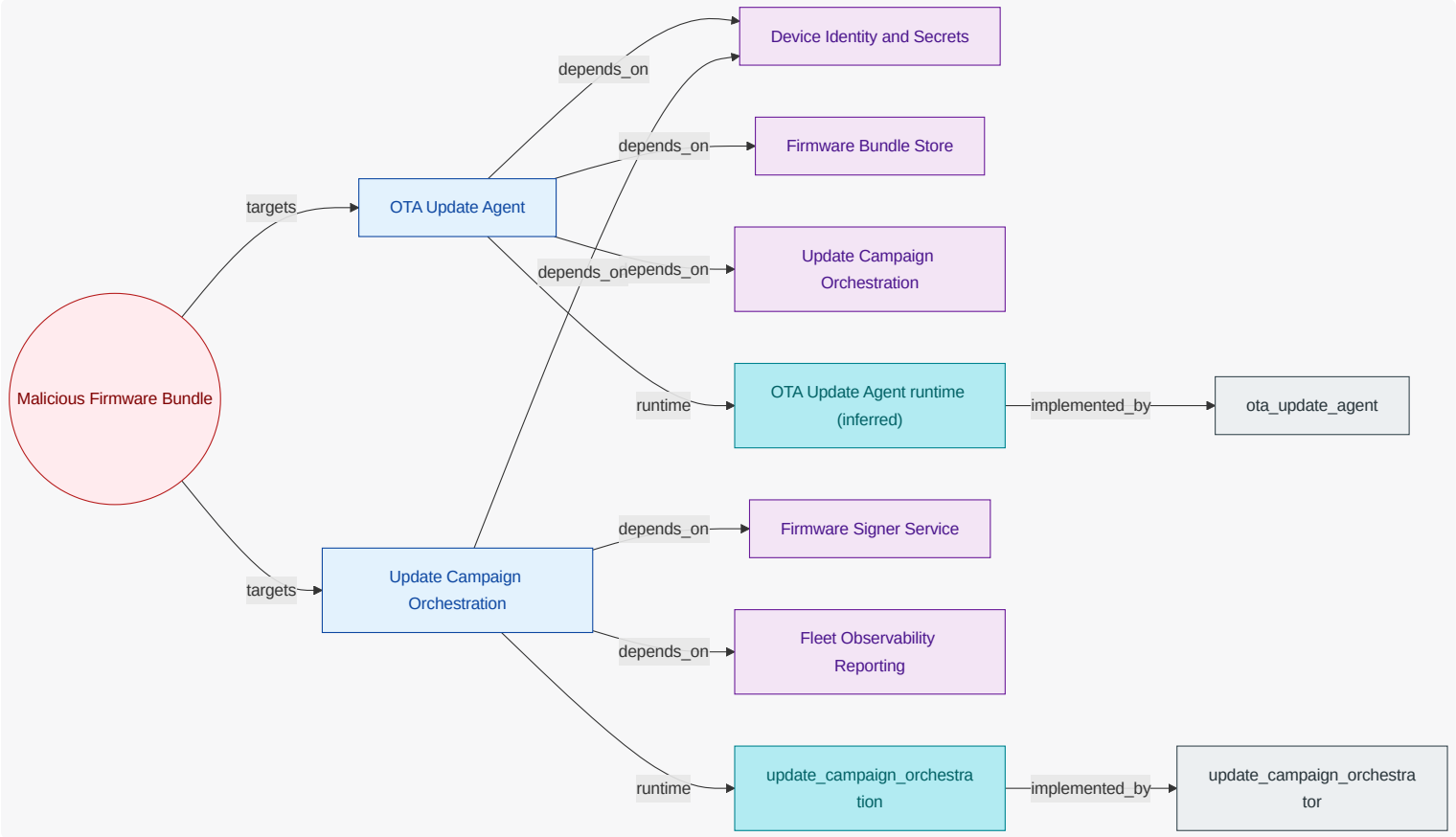
`Cloud Logging Service`, `Device Identity and Secrets`

Security Evidence Present

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

8.8.2. Malicious Firmware Bundle (AV-MALICIOUS-FIRMWARE-BUNDLE)

Tampered OTA payload targeting update agent execution.



OTA Update Agent (FU-OTA-UPDATE-AGENT)

Enforces `signature valid`ation and rejects untrusted `firmware bundles`.

Attack Surface

`Device Identity and Secrets`, `Firmware Bundle Store`, `Update Campaign Orchestration`

Security Evidence Present

code modules: `ota_update_agent`

Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Enforces signed-firmware policy and campaign authorization boundaries.

Attack Surface

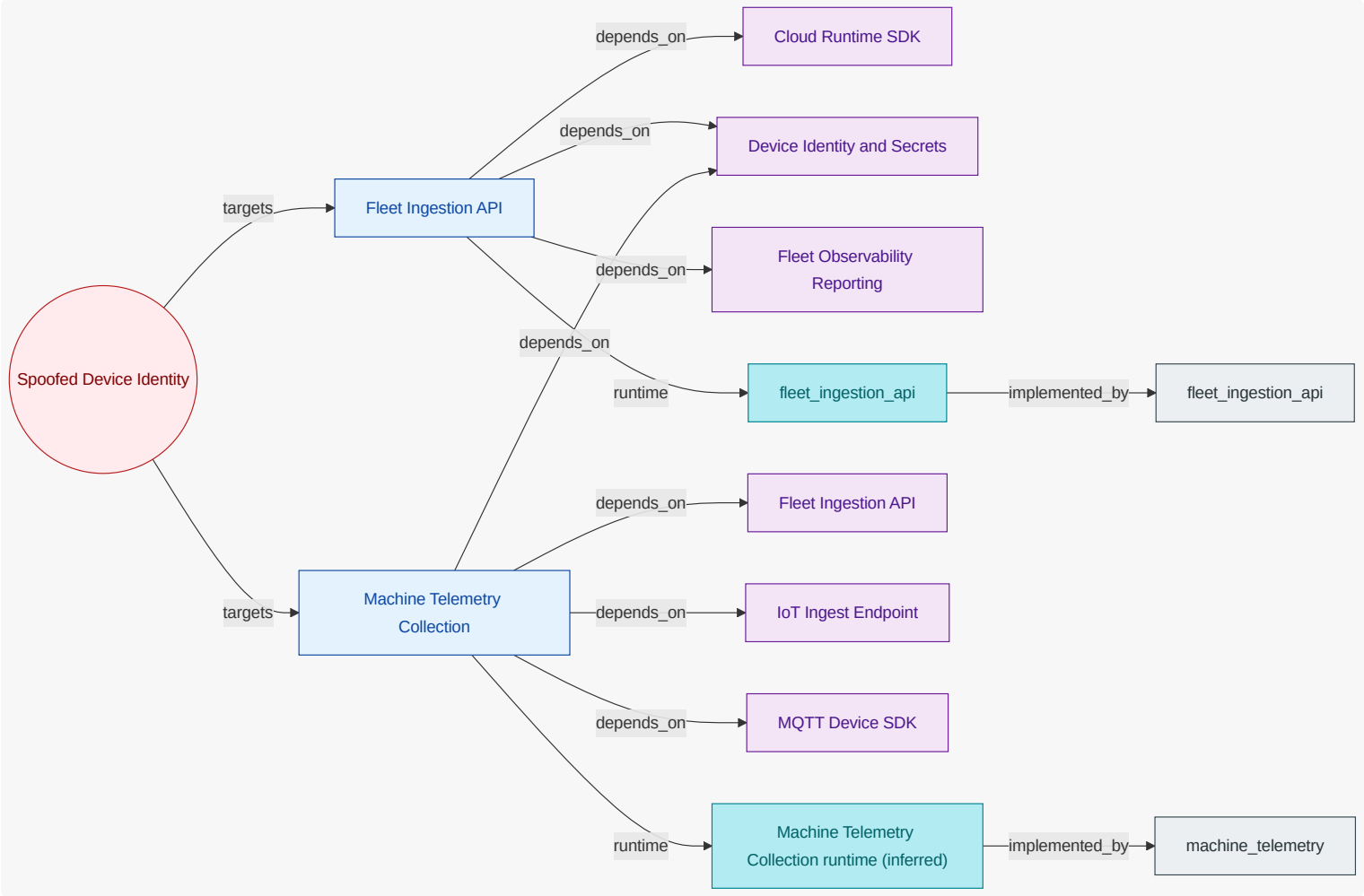
`Device Identity and Secrets`, `Firmware Signer Service`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `update_campaign_orchestration` | code modules: `update_campaign_orchestrator`

8.8.3. `Spoofed Device Identity` (AV-SPOOFED-DEVICE-IDENTITY)

Unauthorized identity attempting `telemetry`/command access.



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

Attack Surface

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Protects `telemetry` authenticity and defends against `replay abuse` patterns.

Attack Surface

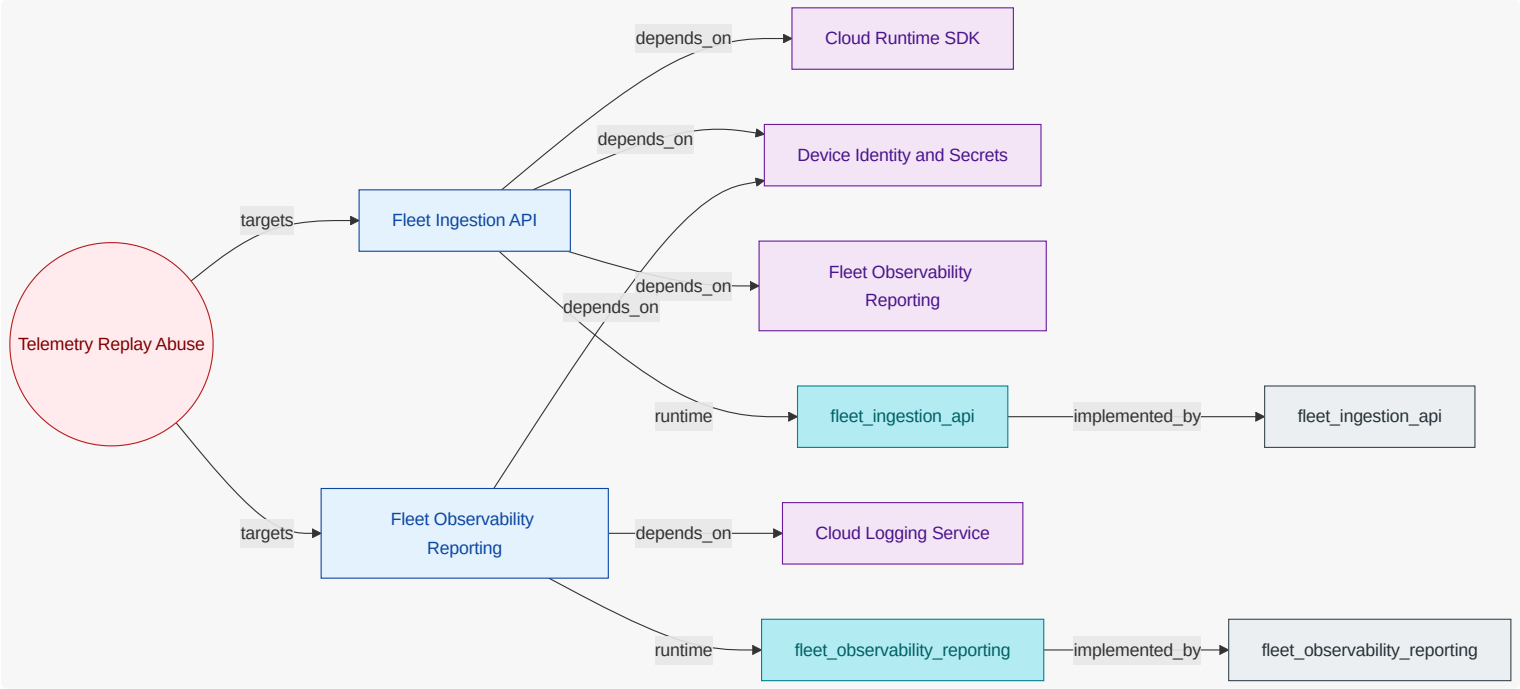
`Device Identity and Secrets`, `Fleet Ingestion API`, `IoT Ingest Endpoint`, `MQTT Device SDK`

Security Evidence Present

code modules: `machine_telemetry`

8.8.4. `Telemetry Replay Abuse` (AV-TELEMETRY-REPLAY-ABUSE)

Replayed `telemetry` intended to overload ingestion and corrupt insights.



Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

Attack Surface

`Cloud Runtime SDK`, `Device Identity and Secrets`, `Fleet Observability Reporting`

Security Evidence Present

runtime: `fleet_ingestion_api` | code modules: `fleet_ingestion_api`

Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

Attack Surface

`Cloud Logging Service`, `Device Identity and Secrets`

Security Evidence Present

runtime: `fleet_observability_reporting` | code modules: `fleet_observability_reporting`

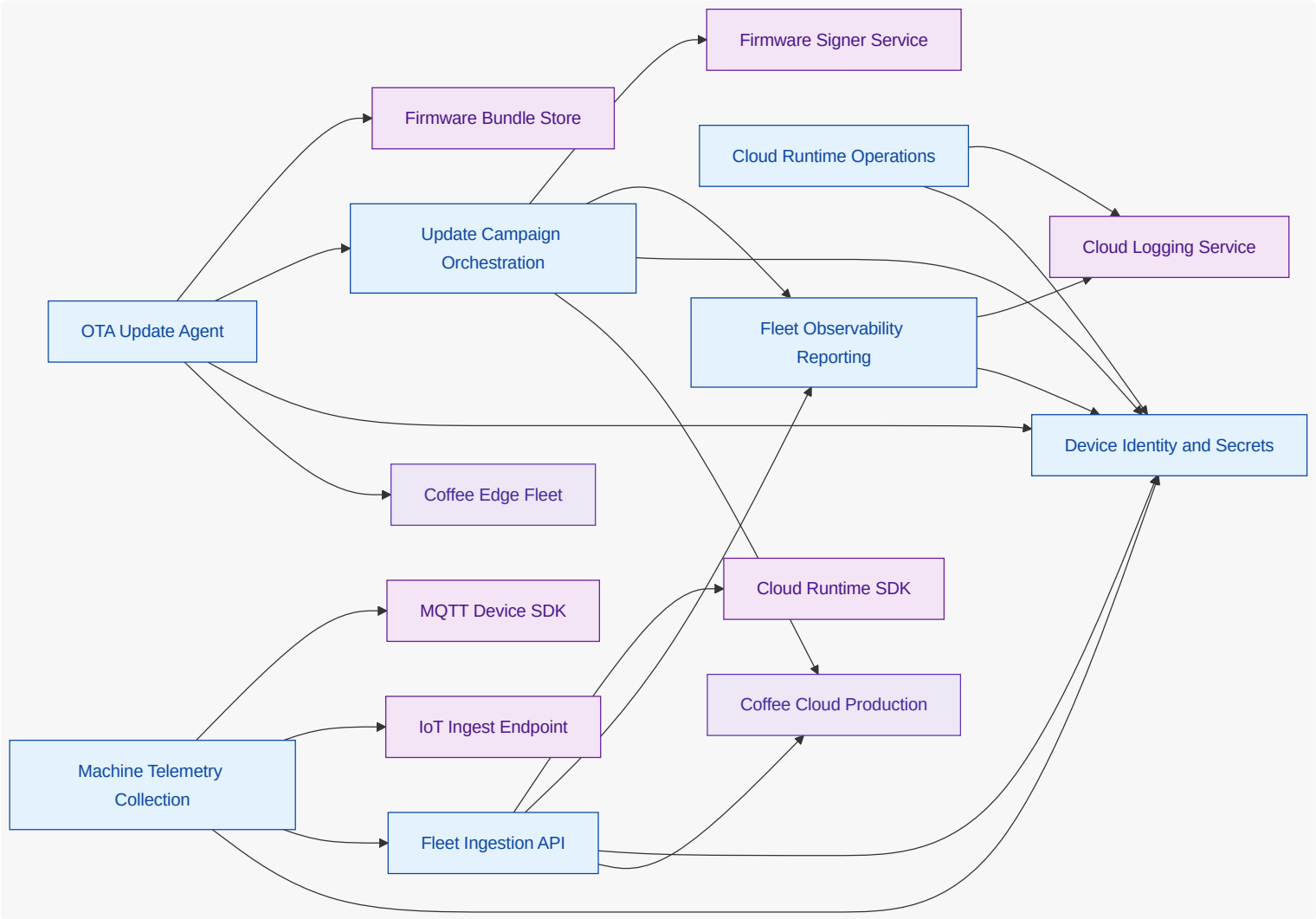
9. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

9.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

9.2. View-Scoped Projection



Node ID	Kind	Label
`DEP-COFFEE-CLOUD-PROD`	deployment_target	`Coffee Cloud Production`
`DEP-COFFEE-EDGE-FLEET`	deployment_target	`Coffee Edge Fleet`

Node ID	Kind	Label
`FU-CLOUD-RUNTIME-OPERATIONS`	functional_unit	`Cloud Runtime Operations`
`FU-DEVICE-IDENTITY-SECRETS`	functional_unit	`Device Identity and Secrets`
`FU-FLEET-INGESTION-API`	functional_unit	`Fleet Ingestion API`
`FU-FLEET-OBSERVABILITY-REPORTING`	functional_unit	`Fleet Observability Reporting`
`FU-MACHINE-TELEMETRY-COLLECTION`	functional_unit	`Machine Telemetry Collection`
`FU-OTA-UPDATE-AGENT`	functional_unit	`OTA Update Agent`
`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	functional_unit	`Update Campaign Orchestration`
`REF-CLOUD-LOGGING-SERVICE`	referenced_element	`Cloud Logging Service`
`REF-CLOUD-RUNTIME-SDK`	referenced_element	`Cloud Runtime SDK`
`REF-FIRMWARE-BUNDLE-STORE`	referenced_element	`Firmware Bundle Store`
`REF-FIRMWARE-SIGNER-SERVICE`	referenced_element	`Firmware Signer Service`
`REF-IOT-INGEST-ENDPOINT`	referenced_element	`IoT Ingest Endpoint`
`REF-MQTT-DEVICE-SDK`	referenced_element	`MQTT Device SDK`

Mapping Type	From	To
depends_on	`FU-CLOUD-RUNTIME-OPERATIONS`	`FU-DEVICE-IDENTITY-SECRETS`
depends_on	`FU-CLOUD-RUNTIME-OPERATIONS`	`REF-CLOUD-LOGGING-SERVICE`
depends_on	`FU-FLEET-INGESTION-API`	`FU-DEVICE-IDENTITY-SECRETS`
depends_on	`FU-FLEET-INGESTION-API`	`FU-FLEET-OBSERVABILITY-REPORTING`

Mapping Type	From	To
depends_on	`FU-FLEET-INGESTION-API`	`REF-CLOUD-RUNTIME-SDK`
depends_on	`FU-FLEET-OBSERVABILITY-REPORTING`	`FU-DEVICE-IDENTITY-SECRETS`
depends_on	`FU-FLEET-OBSERVABILITY-REPORTING`	`REF-CLOUD-LOGGING-SERVICE`
depends_on	`FU-MACHINE-TELEMETRY-COLLECTION`	`FU-DEVICE-IDENTITY-SECRETS`
depends_on	`FU-MACHINE-TELEMETRY-COLLECTION`	`FU-FLEET-INGESTION-API`
depends_on	`FU-MACHINE-TELEMETRY-COLLECTION`	`REF-IOT-INGEST-ENDPOINT`
depends_on	`FU-MACHINE-TELEMETRY-COLLECTION`	`REF-MQTT-DEVICE-SDK`
depends_on	`FU-OTA-UPDATE-AGENT`	`FU-DEVICE-IDENTITY-SECRETS`
depends_on	`FU-OTA-UPDATE-AGENT`	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`
depends_on	`FU-OTA-UPDATE-AGENT`	`REF-FIRMWARE-BUNDLE-STORE`
depends_on	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`FU-DEVICE-IDENTITY-SECRETS`
depends_on	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`FU-FLEET-OBSERVABILITY-REPORTING`
depends_on	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`REF-FIRMWARE-SIGNER-SERVICE`
deployed_to	`FU-FLEET-INGESTION-API`	`DEP-COFFEE-CLOUD-PROD`
deployed_to	`FU-OTA-UPDATE-AGENT`	`DEP-COFFEE-EDGE-FLEET`
deployed_to	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`DEP-COFFEE-CLOUD-PROD`

9.3. Coverage Gaps

Coverage summary: 7/7 units have direct evidence coverage in this view.

- `Device Identity and Secrets` has no direct inferred runtime evidence yet.
- `Machine Telemetry Collection` has no direct inferred runtime evidence yet.
- `OTA Update Agent` has no direct inferred runtime evidence yet.

9.4. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

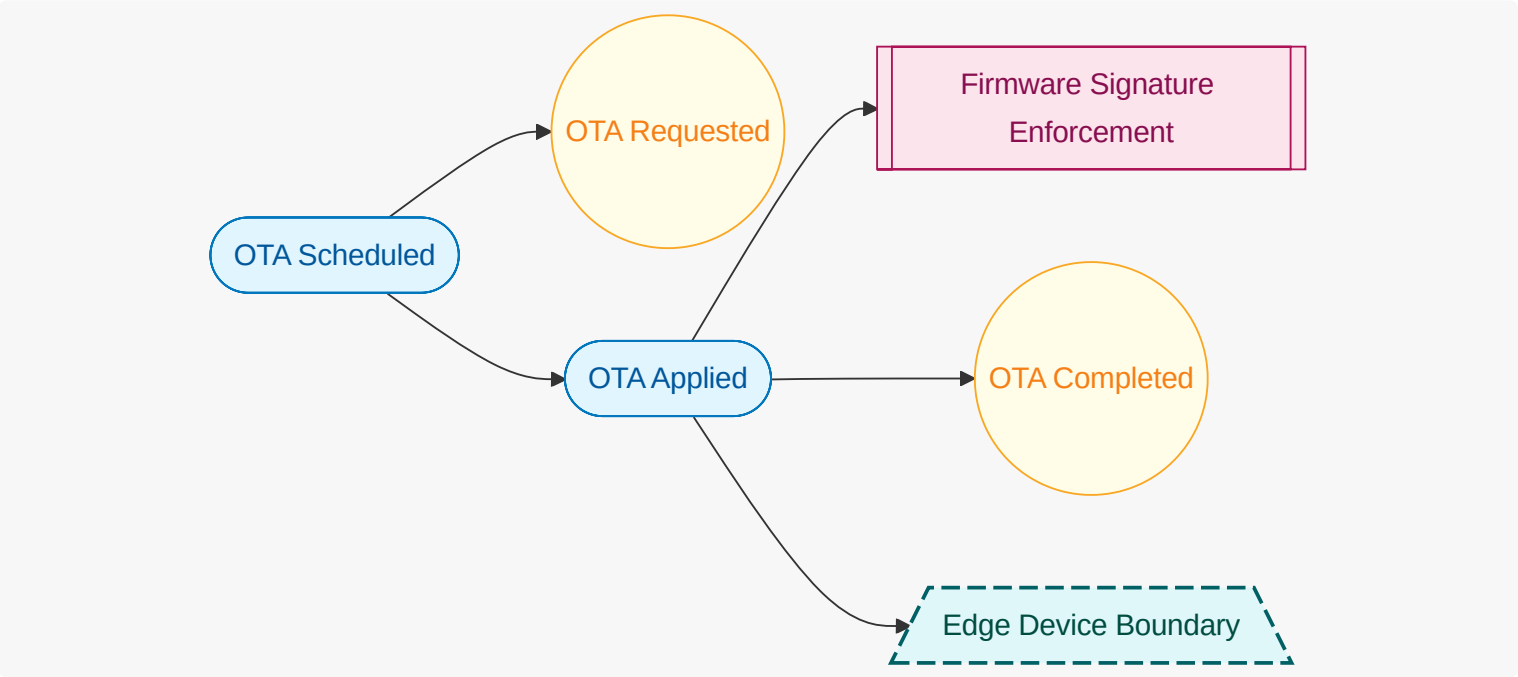
10. State Lifecycle View

Show workflow `state` transitions, ownership of transitions, and key exceptional paths.

10.1. What This View Answers

- What `states` exist and which `events` trigger transitions?
- Which unit owns each `state` transition?
- Where do automatic and manual flows diverge?

10.2. View-Scoped Projection



Node ID	Kind	Label
`CTRL-COFFEE-FIRMWARE-SIGNATURE`	`control`	`Firmware Signature Enforcement`
`EVT-COFFEE-OTA-COMPLETED`	`event`	`OTA Completed`
`EVT-COFFEE-OTA-REQUESTED`	`event`	`OTA Requested`
`STATE-COFFEE-OTA-APPLIED`	`state`	`OTA Applied`
`STATE-COFFEE-OTA-SCHEDULED`	`state`	`OTA Scheduled`
`TB-COFFEE-EDGE-DEVICE`	trust_boundary	`Edge Device Boundary`

Mapping Type	From	To
guarded_by	`STATE-COFFEE-OTA-APPLIED`	`CTRL-COFFEE-FIRMWARE-SIGNATURE`
guarded_by	`STATE-COFFEE-OTA-APPLIED`	`TB-COFFEE-EDGE-DEVICE`
transitions_to	`STATE-COFFEE-OTA-SCHEDULED`	`STATE-COFFEE-OTA-APPLIED`
triggered_by	`STATE-COFFEE-OTA-APPLIED`	`EVT-COFFEE-OTA-COMPLETED`
triggered_by	`STATE-COFFEE-OTA-SCHEDULED`	`EVT-COFFEE-OTA-REQUESTED`

10.3. Coverage Gaps

Coverage summary: No `functional units` are in scope for this view.

- No major coverage gaps detected from current authored and inferred inputs.

10.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view.

10.5. Functional Groups and Units (View Scoped)

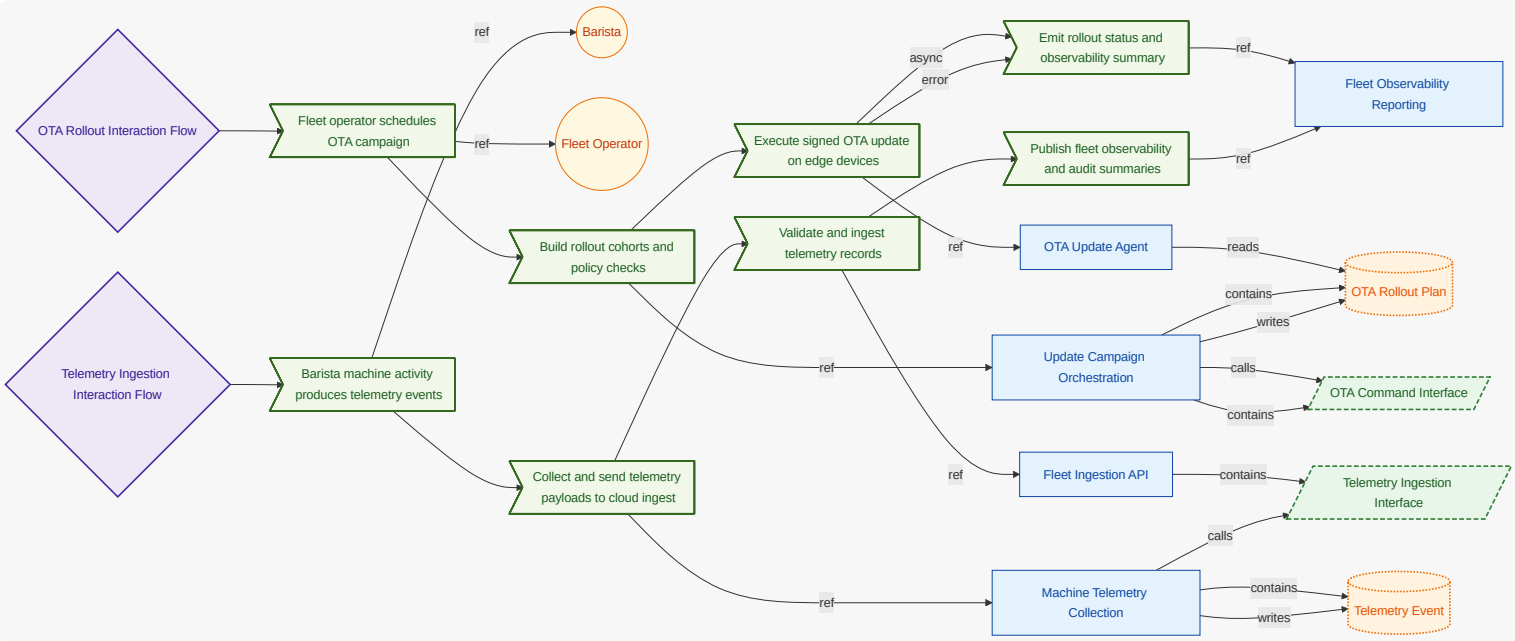
11. Interaction Flow View

Show user-intent-to-outcome causal `flow`, including ordered steps, data movement, and error/async branches.

11.1. What This View Answers

- What happens from user intent to system outcome, step by step?
- What data enters/exits each step and where are decisions made?
- Where are async and error paths and what completes each path?

11.2. View-Scoped Projection



Node ID	Kind	Label
`ACT-BARISTA`	`actor`	`Barista`
`ACT-FLEET-OPERATOR`	`actor`	`Fleet Operator`
`DO-COFFEE-OTA-PLAN`	data_object	`OTA Rollout Plan`
`DO-COFFEE-TELEMETRY-EVENT`	data_object	`Telemetry Event`
`FLOW-COFFEE-OTA-ROLLOUT`	`flow`	`OTA Rollout Interaction Flow`
`FLOW-COFFEE-TELEMETRY-INGEST`	`flow`	`Telemetry Ingestion Interaction Flow`
`FLOW-COFFEE-OTA-ROLLOUT::execute-update`	flow_step	`Execute signed OTA update on edge devices`

Node ID	Kind	Label
`FLOW-COFFEE-OTA-ROLLOUT::operator-schedules-rollout`	flow_step	`Fleet operator schedules OTA campaign`
`FLOW-COFFEE-OTA-ROLLOUT::plan-campaign`	flow_step	`Build rollout cohorts and policy checks`
`FLOW-COFFEE-OTA-ROLLOUT::report-outcome`	flow_step	`Emit rollout status and observability summary`
`FLOW-COFFEE-TELEMETRY-INGEST::capture-machine-telemetry`	flow_step	`Barista machine activity produces telemetry events`
`FLOW-COFFEE-TELEMETRY-INGEST::collect-and-send-telemetry`	flow_step	`Collect and send telemetry payloads to cloud ingest`
`FLOW-COFFEE-TELEMETRY-INGEST::ingest-telemetry`	flow_step	`Validate and ingest telemetry records`
`FLOW-COFFEE-TELEMETRY-INGEST::publish-fleet-observability`	flow_step	`Publish fleet observability and audit summaries`
`FU-FLEET-INGESTION-API`	functional_unit	`Fleet Ingestion API`
`FU-FLEET-OBSERVABILITY-REPORTING`	functional_unit	`Fleet Observability Reporting`
`FU-MACHINE-TELEMETRY-COLLECTION`	functional_unit	`Machine Telemetry Collection`
`FU-OTA-UPDATE-AGENT`	functional_unit	`OTA Update Agent`
`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	functional_unit	`Update Campaign Orchestration`

Node ID	Kind	Label
IF-COFFEE-OTA-COMMAND	interface	OTA Command Interface
IF-COFFEE-TELEMETRY-INGEST	interface	Telemetry Ingestion Interface

Mapping Type	From	To
calls	FU-MACHINE-TELEMETRY-COLLECTION	IF-COFFEE-TELEMETRY-INGEST
calls	FU-UPDATE-CAMPAIGN-ORCHESTRATION	IF-COFFEE-OTA-COMMAND
contains	FU-FLEET-INGESTION-API	IF-COFFEE-TELEMETRY-INGEST
contains	FU-MACHINE-TELEMETRY-COLLECTION	DO-COFFEE-TELEMETRY-EVENT
contains	FU-UPDATE-CAMPAIGN-ORCHESTRATION	DO-COFFEE-OTA-PLAN
contains	FU-UPDATE-CAMPAIGN-ORCHESTRATION	IF-COFFEE-OTA-COMMAND
flow_async	FLOW-COFFEE-OTA-ROLLOUT::execute-update	FLOW-COFFEE-OTA-ROLLOUT::report-outcome
flow_error	FLOW-COFFEE-OTA-ROLLOUT::execute-update	FLOW-COFFEE-OTA-ROLLOUT::report-outcome
flow_next	FLOW-COFFEE-OTA-ROLLOUT	FLOW-COFFEE-OTA-ROLLOUT::operator-schedules-rollout
flow_next	FLOW-COFFEE-OTA-ROLLOUT::operator-schedules-rollout	FLOW-COFFEE-OTA-ROLLOUT::plan-campaign
flow_next	FLOW-COFFEE-OTA-ROLLOUT::plan-campaign	FLOW-COFFEE-OTA-ROLLOUT::execute-update
flow_next	FLOW-COFFEE-TELEMETRY-INGEST	FLOW-COFFEE-TELEMETRY-INGEST::capture-machine-telemetry
flow_next	FLOW-COFFEE-TELEMETRY-INGEST::capture-machine-telemetry	FLOW-COFFEE-TELEMETRY-INGEST::collect-and-send-telemetry
flow_next	FLOW-COFFEE-TELEMETRY-INGEST::collect-and-send-telemetry	FLOW-COFFEE-TELEMETRY-INGEST::ingest-telemetry
flow_next	FLOW-COFFEE-TELEMETRY-INGEST::ingest-telemetry	FLOW-COFFEE-TELEMETRY-INGEST::publish-fleet-observability
flow_ref	FLOW-COFFEE-OTA-ROLLOUT::execute-update	FU-OTA-UPDATE-AGENT
flow_ref	FLOW-COFFEE-OTA-ROLLOUT::operator-schedules-rollout	ACT-FLEET-OPERATOR
flow_ref	FLOW-COFFEE-OTA-ROLLOUT::plan-campaign	FU-UPDATE-CAMPAIGN-ORCHESTRATION
flow_ref	FLOW-COFFEE-OTA-ROLLOUT::report-outcome	FU-FLEET-OBSERVABILITY-REPORTING

Mapping Type	From	To
flow_ref	`FLOW-COFFEE-TELEMETRY-INGEST::capture-machine-telemetry`	`ACT-BARISTA`
flow_ref	`FLOW-COFFEE-TELEMETRY-INGEST::collect-and-send-telemetry`	`FU-MACHINE-TELEMETRY-COLLECTION`
flow_ref	`FLOW-COFFEE-TELEMETRY-INGEST::ingest-telemetry`	`FU-FLEET-INGESTION-API`
flow_ref	`FLOW-COFFEE-TELEMETRY-INGEST::publish-fleet-observability`	`FU-FLEET-OBSERVABILITY-REPORTING`
reads	`FU-OTA-UPDATE-AGENT`	`DO-COFFEE-OTA-PLAN`
writes	`FU-MACHINE-TELEMETRY-COLLECTION`	`DO-COFFEE-TELEMETRY-EVENT`
writes	`FU-UPDATE-CAMPAIGN-ORCHESTRATION`	`DO-COFFEE-OTA-PLAN`

11.3. Coverage Gaps

Coverage summary: 5/5 units have direct evidence coverage in this view.

- No major coverage gaps detected from current authored and inferred inputs.

11.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view.

11.5. Functional Groups and Units (View Scoped)

11.6. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

Requirement	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING	FU-MACHINE-TELEMETRY-COLLECTION	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
REQ-COF-001			X		X		
REQ-COF-002			X	X			
REQ-COF-003						X	X
REQ-COF-004		X				X	
REQ-COF-005				X		X	X
REQ-COF-006			X		X		
REQ-COF-007		X		X			

Requirement	FU-CLOUD-RUNTIME-OPERATIONS	FU-DEVICE-IDENTITY-SECRETS	FU-FLEET-INGESTION-API	FU-FLEET-OBSERVABILITY-REPORTING	FU-MACHINE-TELEMETRY-COLLECTION	FU-OTA-UPDATE-AGENT	FU-UPDATE-CAMPAIGN-ORCHESTRATION
REQ-COF-008	X		X	X			
REQ-COF-009			X	X			
REQ-COF-010				X		X	X
REQ-COF-011			X		X		
REQ-COF-012	X		X	X			

11.7. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794`	`REQ-COF-001`	pass	test-results/contract-telemetry.json ingest payload accepted with `machine identity` and `telemetry` fields
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`	`REQ-COF-002`	not-run	none No parsed test-result artifact for this `requirement` yet.
`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`	`REQ-COF-003`	not-run	none No parsed test-result artifact for this `requirement` yet.
`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`	`REQ-COF-003`	pass	test-results/unit-ota.xml
`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`	`REQ-COF-004`	pass	test-results/unit-ota.xml
`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`	`REQ-COF-005`	partial	test-results/ota-rollback-report.md
`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`	`REQ-COF-007`	pass	test-results/integration-logging.json `audit log record` persisted with redacted `machine identity`

12. Traceability Appendix

12.1. Requirement Details

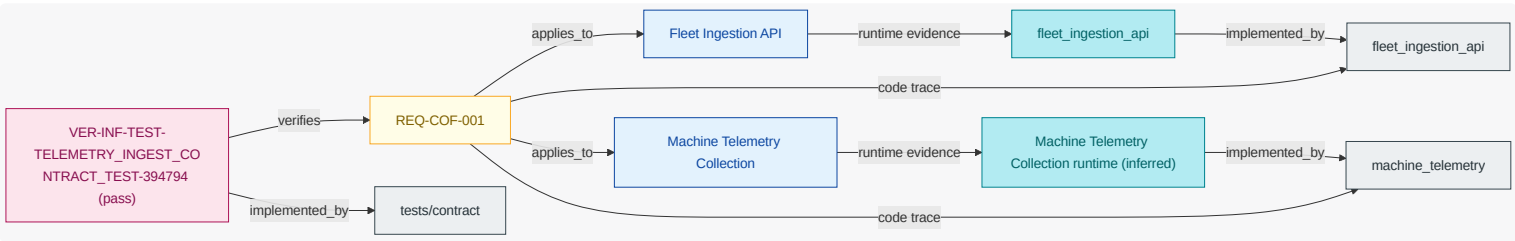
12.1.1. REQ-COF-001

While `coffee machine fleet operation is active`, when `brew cycle completed event is received`, `the coffee fleet system` shall publish `brew telemetry` to `fleet ingestion api` for `barista` operations.

> **NOTE** Primary machine-to-cloud `telemetry` contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



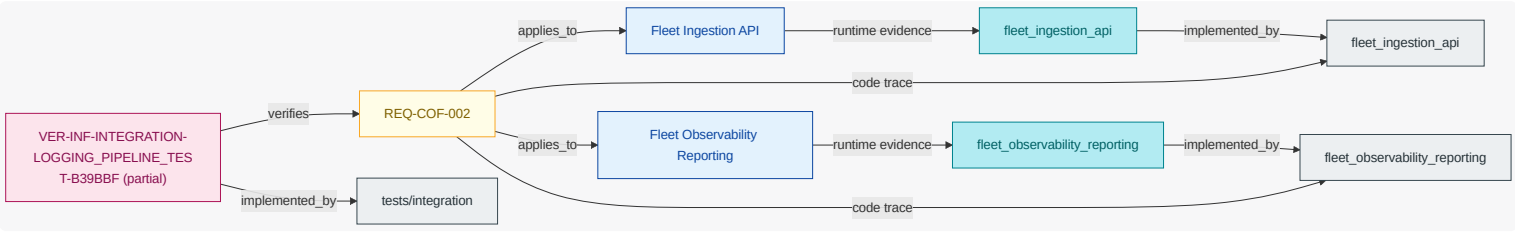
12.1.2. REQ-COF-002

While `coffee machine fleet operation is active`, when `telemetry ingested event is received`, `the coffee fleet system` shall persist fleet `telemetry` metrics for `fleet operator`.

> **NOTE** Operational observability `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



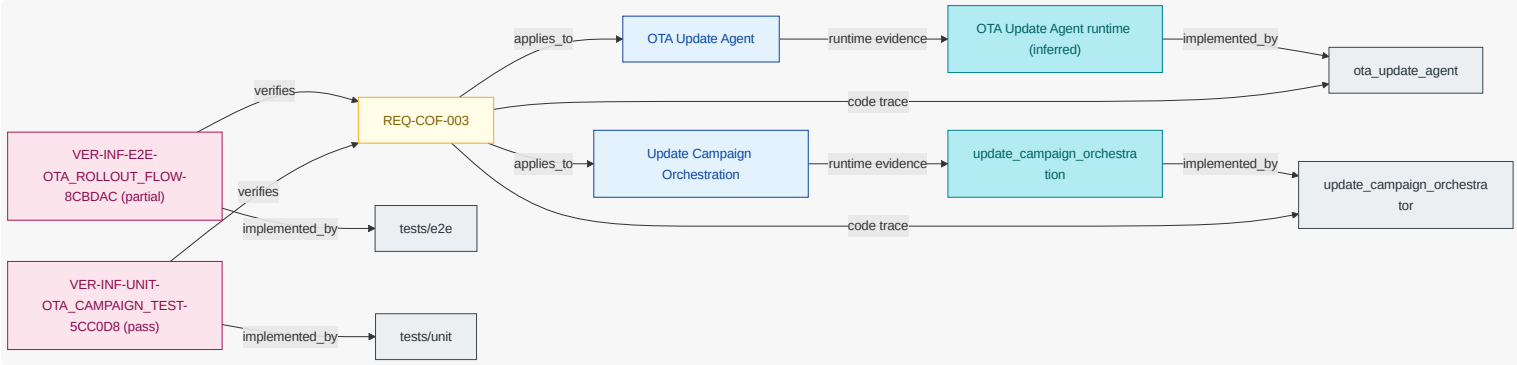
12.1.3. REQ-COF-003

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall distribute `signed firmware bundle` to targeted machine cohort using `ota campaign policy`.

> **NOTE** `OTA rollout` behavior from central `cloud control`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



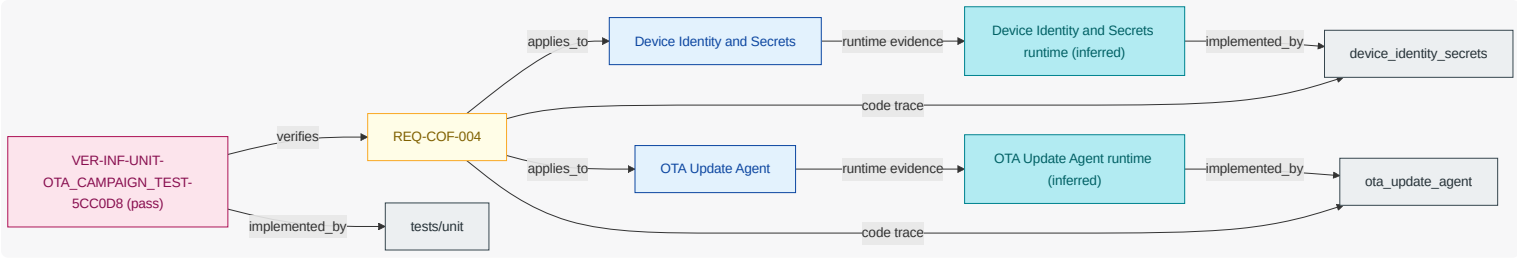
12.1.4. REQ-COF-004

Where `ota campaign is enabled`, when `ota campaign scheduled event is received`, `the coffee fleet system` shall gate `firmware apply` on `firmware signature is valid`.

> **NOTE** Firmware integrity and `trust boundary` `control`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



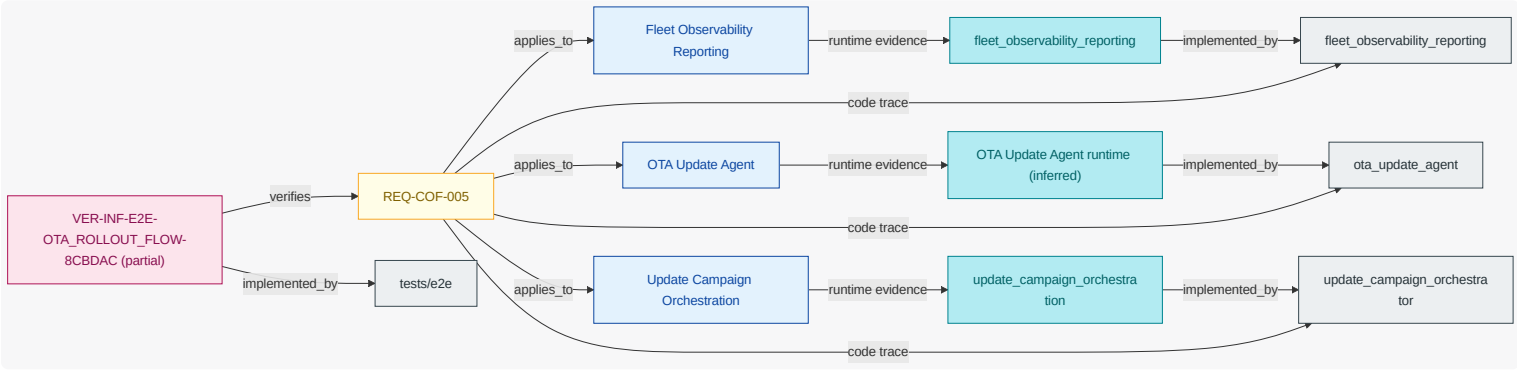
12.1.5. REQ-COF-005

If `firmware validation failed event is received`, then `the coffee fleet system` shall `rollback firmware apply`.

> **NOTE** Rollback and failure signaling path.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



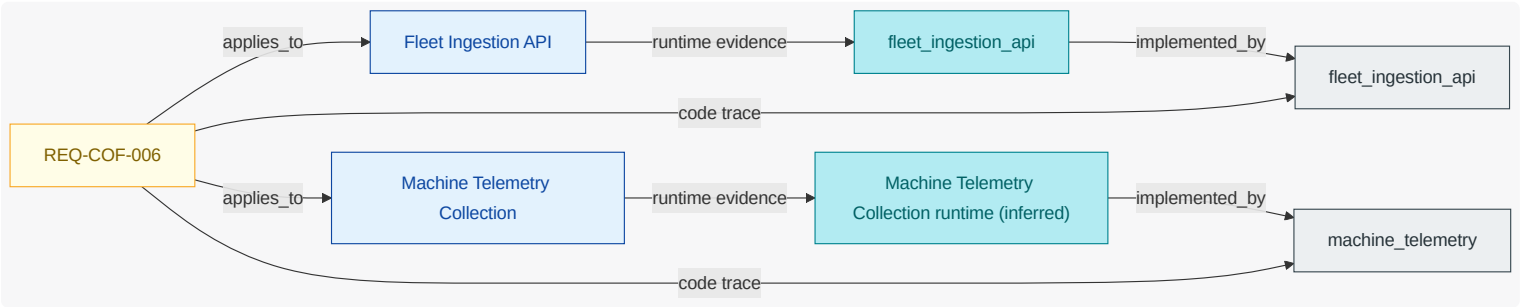
12.1.6. REQ-COF-006

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall queue `telemetry` for retry.

> **NOTE** Resilience for temporary cloud outage.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



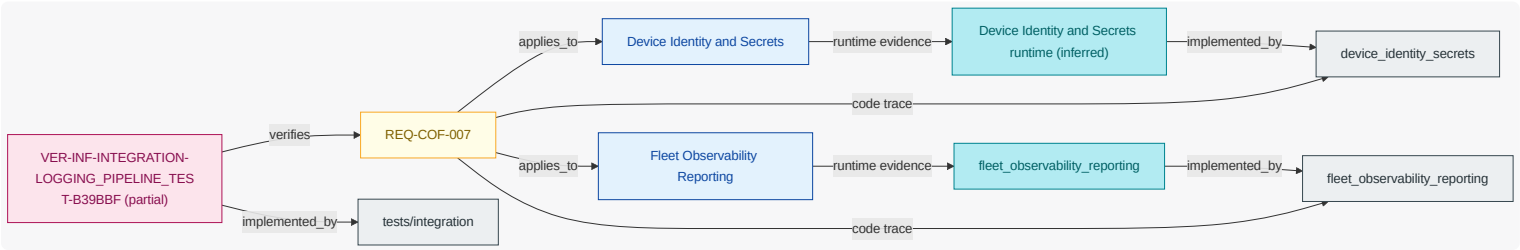
12.1.7. REQ-COF-007

Where `audit logging is enabled`, when `ota status reported event is received`, `the coffee fleet system` shall persist redacted `audit log record` with `machine identity` plus `campaign id`.

> **NOTE** Audit/compliance logging `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



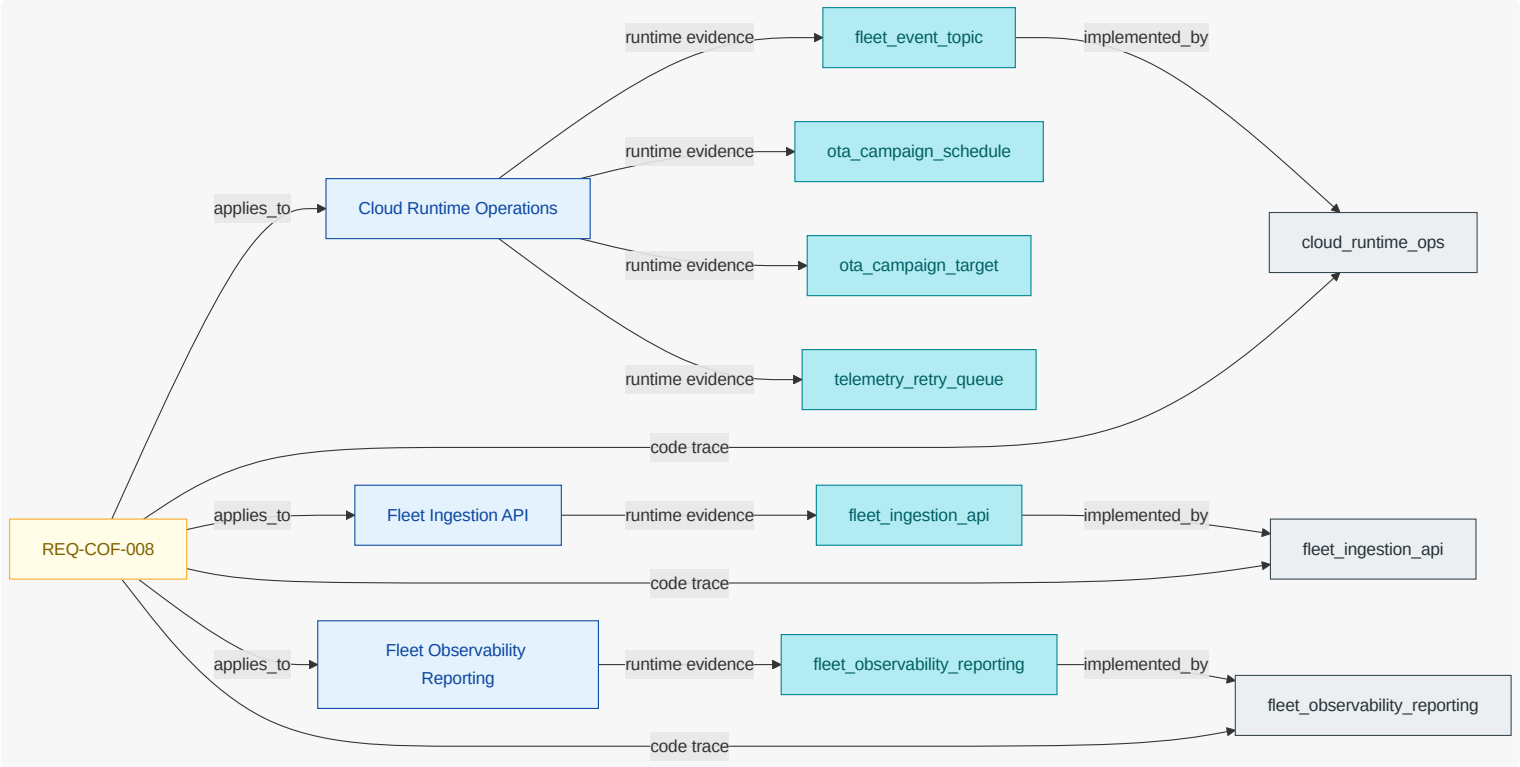
12.1.8. REQ-COF-008

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall throttle duplicate `telemetry`.

> **NOTE** Abuse `control` and security response `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



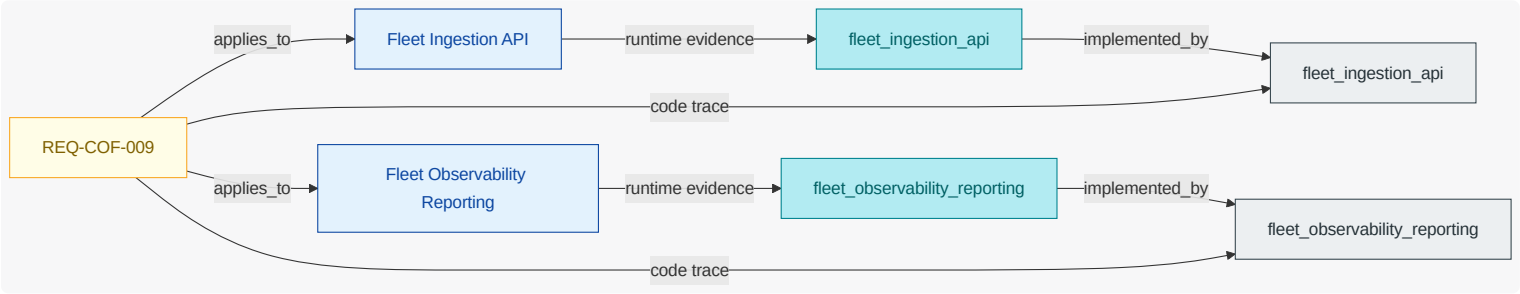
12.1.9. REQ-COF-009

While `coffee machine fleet operation is active`, when `telemetry ingested event is received`, `the coffee fleet system` shall persist `logging records` for `fleet operator`.

> **NOTE** Operational logging persistence `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



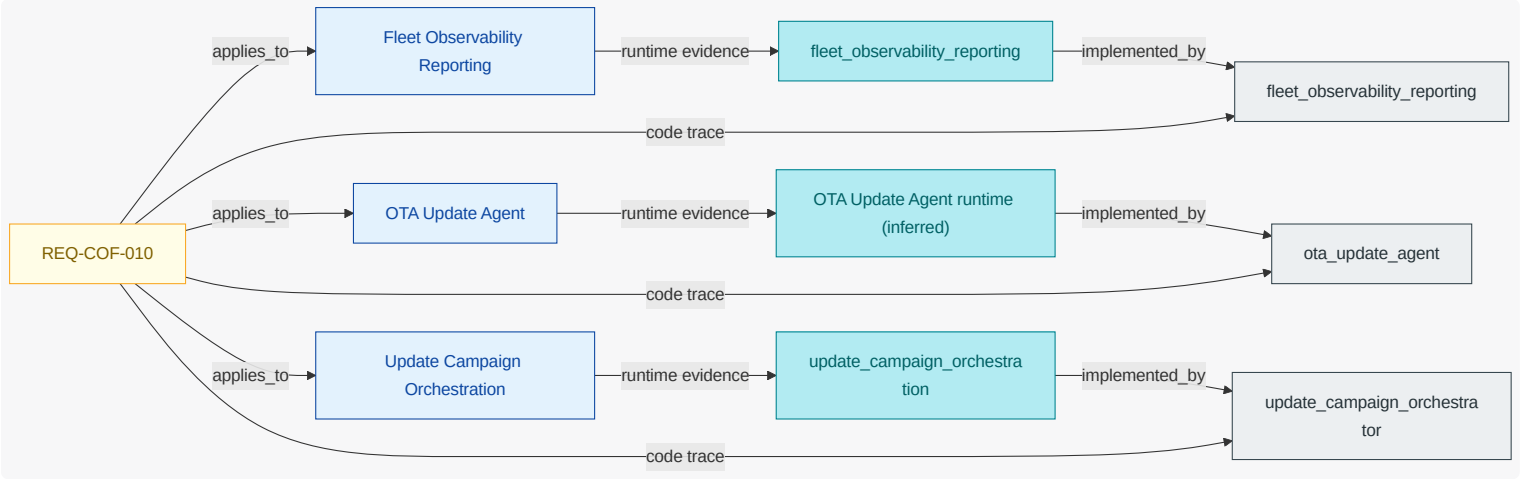
12.1.10. REQ-COF-010

If `firmware validation failed event is received`, then `the coffee fleet system` shall report `ota status` to `platform operator`.

> **NOTE** OTA failure status signaling `requirement`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



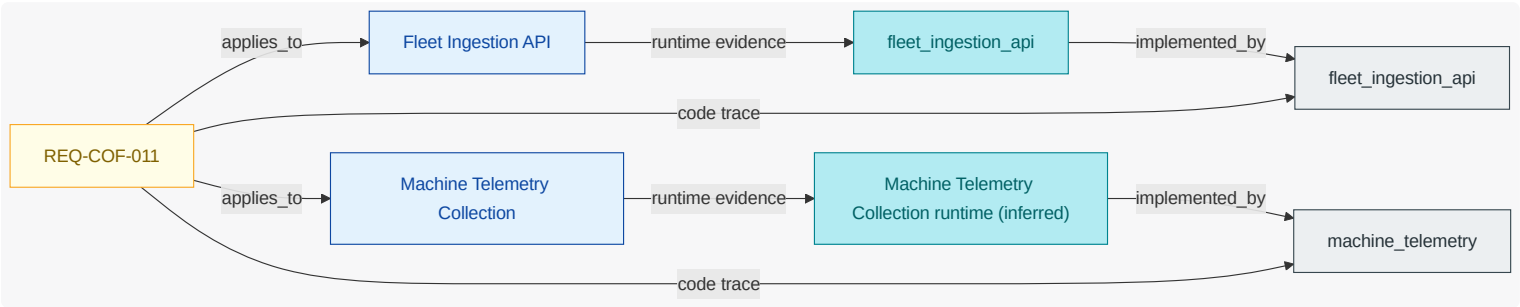
12.1.11. REQ-COF-011

If `cloud ingest endpoint is unavailable`, then `the coffee fleet system` shall flush queued `telemetry` when `normal mode is active`.

> **NOTE** Queued `telemetry` drain behavior after endpoint recovery.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



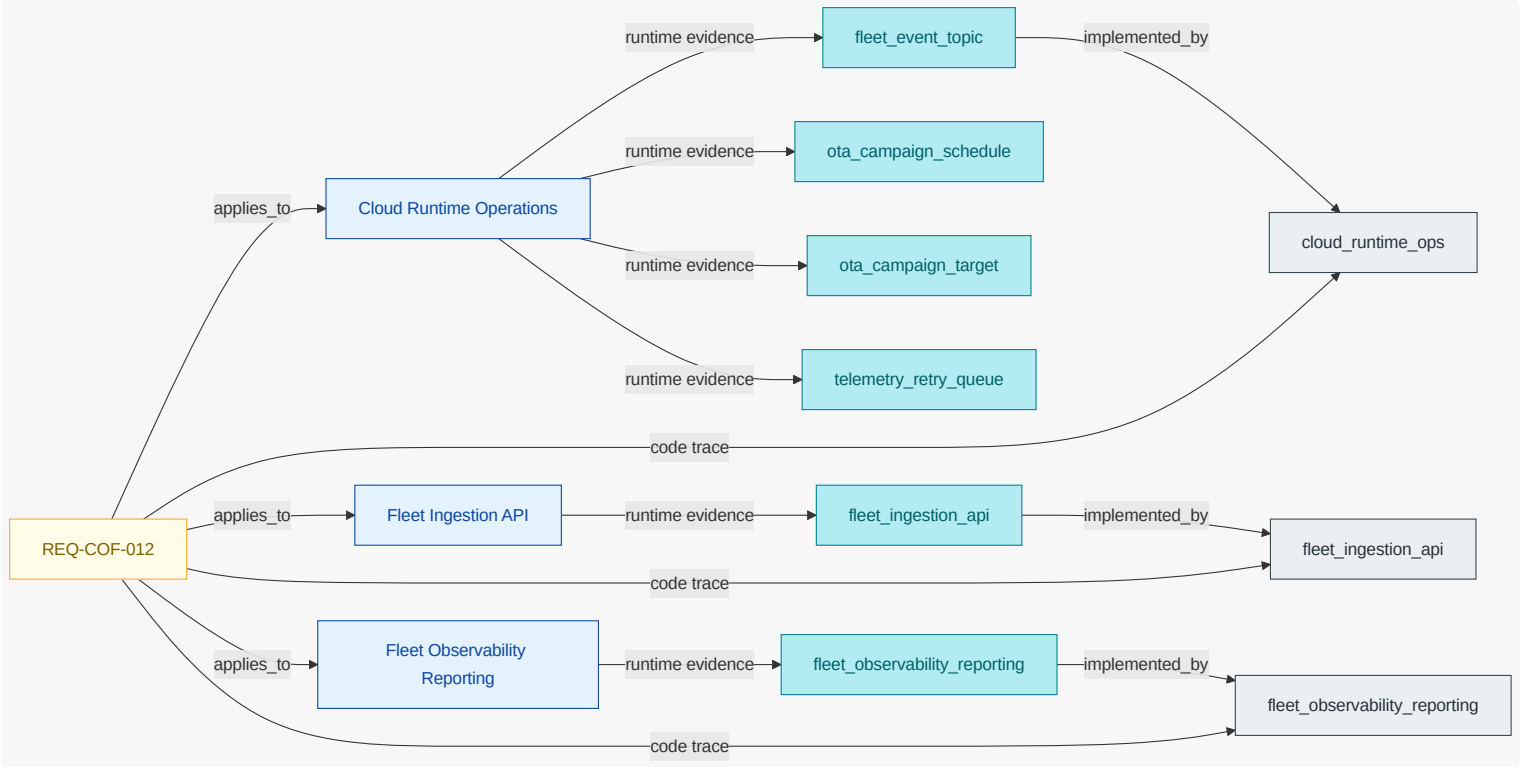
12.1.12. REQ-COF-012

If `telemetry replay abuse is detected`, then `the coffee fleet system` shall notify `security analyst`.

> **NOTE** Security notification `requirement` for `replay abuse` `events`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



12.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

12.2.1. Ota Rollout Flow

Field	Value
ID	`VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC`
Kind	e2e
Status	partial
Verifies Requirements	`REQ-COF-003`, `REQ-COF-005`
Test Code Elements	tests/e2e/ota_rollout_flow.yaml
Derived Functional Owners	`FU-FLEET-OBSERVABILITY-REPORTING`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION`
Result Summary	not-run:1, partial:1
Evidence	test-results/ota-rollback-report.md, tests/e2e/ota_rollout_flow.yaml
Notes	runs fleet `OTA rollout` scenario including scheduling, deployment, and `telemetry` confirmation

12.2.2. Logging Pipeline Test

Field	Value
ID	`VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF`

Field	Value
Kind	integration
Status	partial
Verifies Requirements	`REQ-COF-002`, `REQ-COF-007`
Test Code Elements	tests/integration/logging_pipeline_test.ts
Derived Functional Owners	`FU-DEVICE-IDENTITY-SECRETS`, `FU-FLEET-INGESTION-API`, `FU-FLEET-OBSERVABILITY-REPORTING`
Result Summary	not-run:1, pass:1
Evidence	test-results/integration-logging.json, tests/integration/logging_pipeline_test.ts
Notes	validates `telemetry` logging pipeline persistence and operational reporting signals

12.2.3. Telemetry Ingest Contract Test

Field	Value
ID	`VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794`
Kind	test
Status	pass
Verifies Requirements	`REQ-COF-001`
Test Code Elements	tests/contract/telemetry_ingest_contract_test.go
Derived Functional Owners	`FU-FLEET-INGESTION-API`, `FU-MACHINE-TELEMETRY-COLLECTION`
Result Summary	pass:1
Evidence	test-results/contract-telemetry.json, tests/contract/telemetry_ingest_contract_test.go
Notes	validates `telemetry` ingest contract fields and accepted request semantics

12.2.4. Ota Campaign Test

Field	Value
ID	`VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8`
Kind	unit
Status	pass
Verifies Requirements	`REQ-COF-003`, `REQ-COF-004`
Test Code Elements	tests/unit/ota_campaign_test.go

Field	Value
Derived Functional Owners	`FU-DEVICE-IDENTITY-SECRETS`, `FU-OTA-UPDATE-AGENT`, `FU-UPDATE-CAMPAIGN-ORCHESTRATION`
Result Summary	pass:2
Evidence	test-results/unit-ota.xml, tests/unit/ota_campaign_test.go
Notes	verifies OTA campaign `signature valid`ation and rollout trigger behavior

13. Reference Index

13.1. Authored References

CTRL-COFFEE-DEVICE-IDENTITY

category=identity-access; description=Require authenticated device identity for telemetry and OTA command channels.

Key	Value
Kind	Control
Name	Device Identity Validation
Mentioned In	`Security View`

CTRL-COFFEE-FIRMWARE-SIGNATURE

category=integrity; description=Require signed firmware bundles before applying OTA updates.

Key	Value
Kind	Control
Name	Firmware Signature Enforcement
Mentioned In	`Architecture Intent View`, `Security View`, `State Lifecycle View`

DO-COFFEE-OTA-PLAN

termRef=DATA-OTA-PLAN; schemaRef=schemas/ota-plan.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	OTA Rollout Plan
Mentioned In	`Architecture Intent View`, `Communication View`, `Interaction Flow View`

DO-COFFEE-TELEMETRY-EVENT

termRef=DATA-TELEMETRY-EVENT; schemaRef=schemas/telemetry-event.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	Telemetry Event
Mentioned In	`Architecture Intent View`, `Communication View`, `Interaction Flow View`

DEP-COFFEE-CLOUD-PROD

environment=prod; region=us-east-1; account=cloud; cluster=coffee-cloud; namespace=coffee; trustZone=app

Key	Value
Kind	Deployment Target
Name	Coffee Cloud Production
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`

DEP-COFFEE-EDGE-FLEET

environment=edge; region=global; account=device; cluster=edge-fleet; namespace=devices; trustZone=edge

Key	Value
Kind	Deployment Target
Name	Coffee Edge Fleet
Mentioned In	`Architecture Intent View`, `Deployment View`, `Traceability View`

EVT-COFFEE-OTA-COMPLETED

OTA rollout completed for cohort.

Key	Value
Kind	Event
Name	OTA Completed
Mentioned In	`Architecture Intent View`, `State Lifecycle View`

EVT-COFFEE-OTA-REQUESTED

OTA campaign request accepted.

Key	Value
Kind	Event
Name	OTA Requested
Mentioned In	`Architecture Intent View`, `State Lifecycle View`

FLOW-COFFEE-OTA-ROLLOUT

entry=operator-schedules-rollout; exits=report-outcome; steps=4

Key	Value
Kind	Flow
Name	OTA Rollout Interaction Flow
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-TELEMETRY-INGEST

entry=capture-machine-telemetry; exits=publish-fleet-observability; steps=4

Key	Value
Kind	Flow
Name	Telemetry Ingestion Interaction Flow
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-OTA-ROLLOUT::execute-update

kind=system_action; ref=FU-OTA-UPDATE-AGENT; action=Execute signed OTA update on edge devices; dataIn=ota_rollout_plan; dataOut=; async=true

Key	Value
Kind	Flow Step
Name	Execute signed OTA update on edge devices
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-OTA-ROLLOUT::operator-schedules-rollout

kind=user_action; ref=ACT-FLEET-OPERATOR; action=Fleet operator schedules OTA campaign; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Fleet operator schedules OTA campaign
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-OTA-ROLLOUT::plan-campaign

kind=system_action; ref=FU-UPDATE-CAMPAIGN-ORCHESTRATION; action=Build rollout cohorts and policy checks; dataIn=; dataOut=ota_rollout_plan; async=false

Key	Value
Kind	Flow Step
Name	Build rollout cohorts and policy checks
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-OTA-ROLLOUT::report-outcome

kind=system_action; ref=FU-FLEET-OBSERVABILITY-REPORTING; action=Emit rollout status and observability summary; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Emit rollout status and observability summary
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-TELEMETRY-INGEST::capture-machine-telemetry

kind=user_action; ref=ACT-BARISTA; action=Barista machine activity produces telemetry events; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Barista machine activity produces telemetry events
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-TELEMETRY-INGEST::collect-and-send-telemetry

kind=system_action; ref=FU-MACHINE-TELEMETRY-COLLECTION; action=Collect and send telemetry payloads to cloud ingest; dataIn=; dataOut=telemetry_event; async=false

Key	Value
Kind	Flow Step
Name	Collect and send telemetry payloads to cloud ingest
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-TELEMETRY-INGEST::ingest-telemetry

kind=system_action; ref=FU-FLEET-INGESTION-API; action=Validate and ingest telemetry records; dataIn=telemetry_event; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Validate and ingest telemetry records
Mentioned In	`Interaction Flow View`

FLOW-COFFEE-TELEMETRY-INGEST::publish-fleet-observability

kind=system_action; ref=FU-FLEET-OBSERVABILITY-REPORTING; action=Publish fleet observability and audit summaries; dataIn=telemetry_event; dataOut=; async=false

Key	Value
Kind	Flow Step

Key	Value
Name	Publish fleet observability and audit summaries
Mentioned In	`Interaction Flow View`

IF-COFFEE-OTA-COMMAND

protocol=mqtt; endpoint=mqtt://devices/ota/commands; schemaRef=schemas/ota-command.json; owner=FU-UPDATE-CAMPAIGN-ORCHESTRATION

Key	Value
Kind	Interface
Name	OTA Command Interface
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Interaction Flow View`

IF-COFFEE-TELEMETRY-INGEST

protocol=mqtt; endpoint=mqtt://ingest/topics/telemetry; schemaRef=schemas/telemetry.json; owner=FU-FLEET-INGESTION-API

Key	Value
Kind	Interface
Name	Telemetry Ingestion Interface
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Interaction Flow View`

REQ-COF-001

While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api for barista operations.

Key	Value
Kind	Requirement
Name	REQ-COF-001
Mentioned In	`Architecture Intent View`, `Deployment View`, `Interaction Flow View`, `Telemetry Ingest Contract Test`

REQ-COF-002

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics for fleet operator.

Key	Value
Kind	Requirement
Name	REQ-COF-002

Key	Value
Mentioned In	`Architecture Intent View`, `Deployment View`, `Interaction Flow View`, `Logging Pipeline Test`

REQ-COF-003

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort using ota campaign policy.

Key	Value
Kind	Requirement
Name	REQ-COF-003
Mentioned In	`Architecture Intent View`, `Deployment View`, `Interaction Flow View`, `Ota Rollout Flow`, `Ota Campaign Test`

REQ-COF-004

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall gate firmware apply on firmware signature is valid.

Key	Value
Kind	Requirement
Name	REQ-COF-004
Mentioned In	`Architecture Intent View`, `Deployment View`, `Interaction Flow View`, `Ota Campaign Test`

REQ-COF-005

If firmware validation failed event is received, then the coffee fleet system shall rollback firmware apply.

Key	Value
Kind	Requirement
Name	REQ-COF-005
Mentioned In	`Architecture Intent View`, `Deployment View`, `Interaction Flow View`, `Ota Rollout Flow`

REQ-COF-006

If cloud ingest endpoint is unavailable, then the coffee fleet system shall queue telemetry for retry.

Key	Value
Kind	Requirement
Name	REQ-COF-006
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-007

Where audit logging is enabled, when ota status reported event is received, the coffee fleet system shall persist redacted audit log record with machine identity plus campaign id.

Key	Value
Kind	Requirement
Name	REQ-COF-007
Mentioned In	`Architecture Intent View`, `Deployment View`, `Interaction Flow View`, `Logging Pipeline Test`

REQ-COF-008

If telemetry replay abuse is detected, then the coffee fleet system shall throttle duplicate telemetry.

Key	Value
Kind	Requirement
Name	REQ-COF-008
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-009

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist logging records for fleet operator.

Key	Value
Kind	Requirement
Name	REQ-COF-009
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-010

If firmware validation failed event is received, then the coffee fleet system shall report ota status to platform operator.

Key	Value
Kind	Requirement
Name	REQ-COF-010
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-011

If cloud ingest endpoint is unavailable, then the coffee fleet system shall flush queued telemetry when normal mode is active.

Key	Value
Kind	Requirement
Name	REQ-COF-011
Mentioned In	`Architecture Intent View`, `Deployment View`

REQ-COF-012

If telemetry replay abuse is detected, then the coffee fleet system shall notify security analyst.

Key	Value
Kind	Requirement
Name	REQ-COF-012
Mentioned In	`Architecture Intent View`, `Deployment View`

STATE-COFFEE-OTA-APPLIED

Firmware update successfully applied to target cohort.

Key	Value
Kind	State
Name	OTA Applied
Mentioned In	`Architecture Intent View`, `State Lifecycle View`

STATE-COFFEE-OTA-SCHEDULED

Campaign prepared and waiting for rollout execution.

Key	Value
Kind	State
Name	OTA Scheduled
Mentioned In	`Architecture Intent View`, `State Lifecycle View`

TB-COFFEE-CLOUD-CONTROL

Boundary between cloud app workloads and privileged platform controls.

Key	Value
Kind	Trust Boundary
Name	Cloud Control Boundary
Mentioned In	`Architecture Intent View`, `Security View`

TB-COFFEE-EDGE-DEVICE

Boundary between managed edge devices and cloud control plane.

Key	Value
Kind	Trust Boundary
Name	Edge Device Boundary
Mentioned In	`Architecture Intent View`, `Security View`, `State Lifecycle View`

13.2. Catalog References

ACT-BARISTA

Operates coffee machines and observes local machine health.

Key	Value
Kind	Catalog Actor
Name	barista
Mentioned In	`Architecture Intent View`, `Communication View`, `Interaction Flow View`, `REQ-COF-001`

ACT-FLEET-OPERATOR

Owns OTA rollout campaigns and fleet-level reliability.

Key	Value
Kind	Catalog Actor
Name	fleet operator
Mentioned In	`Architecture Intent View`, `Communication View`, `Interaction Flow View`, `REQ-COF-002`, `REQ-COF-009`

ACT-PLATFORM-OPERATOR

Owns runtime deployment and operational controls.

Key	Value
Kind	Catalog Actor
Name	platform operator
Mentioned In	`Architecture Intent View`, `Communication View`, `REQ-COF-010`

ACT-SECURITY-ANALYST

Reviews identity, firmware integrity, and logging controls.

Key	Value
Kind	Catalog Actor
Name	security analyst
Mentioned In	`Architecture Intent View`, `Communication View`, `REQ-COF-012`

apply status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	apply status
Mentioned In	n/a

apply/rollback status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	apply/rollback status
Mentioned In	n/a

audit enabled

Feature flag enabling audit-grade log persistence.

Key	Value
Kind	Catalog Alias
Name	audit enabled
Mentioned In	n/a

audit records

Immutable log record for security and compliance analysis.

Key	Value
Kind	Catalog Alias
Name	audit records
Mentioned In	`Architecture Intent View`

audit tampering

Attempts to suppress or alter security/operational log records.

Key	Value
Kind	Catalog Alias
Name	audit tampering
Mentioned In	n/a

brew completed

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Catalog Alias
Name	brew completed
Mentioned In	n/a

brew cycle event

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Catalog Alias
Name	brew cycle event
Mentioned In	n/a

campaign commands

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	campaign commands
Mentioned In	n/a

campaign identifier

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Catalog Alias

Key	Value
Name	campaign identifier
Mentioned In	n/a

campaign ids

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Catalog Alias
Name	campaign ids
Mentioned In	n/a

campaign orchestration

Selects target machine groups and coordinates OTA rollout phases.

Key	Value
Kind	Catalog Alias
Name	campaign orchestration
Mentioned In	n/a

campaign policy

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Alias
Name	campaign policy
Mentioned In	n/a

campaign scheduled

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	campaign scheduled
Mentioned In	n/a

central cloud site

Capability area for fleet ingestion, campaign orchestration, and cloud reporting.

Key	Value
Kind	Catalog Alias
Name	central cloud site
Mentioned In	n/a

cloud sdk

SDK used by cloud units for runtime integrations.

Key	Value
Kind	Catalog Alias
Name	cloud sdk
Mentioned In	n/a

coffee machine edge

Capability area for device-local telemetry and OTA execution behavior.

Key	Value
Kind	Catalog Alias
Name	coffee machine edge
Mentioned In	n/a

device identity

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Catalog Alias
Name	device identity
Mentioned In	n/a

device secrets

Manages device identity material and secure configuration.

Key	Value
Kind	Catalog Alias
Name	device secrets

Key	Value
Mentioned In	n/a

device spoofing

Unauthorized actor impersonates a machine identity.

Key	Value
Kind	Catalog Alias
Name	device spoofing
Mentioned In	n/a

device update agent

Downloads, validates, and applies signed firmware updates on machines.

Key	Value
Kind	Catalog Alias
Name	device update agent
Mentioned In	n/a

endpoint outage

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Catalog Alias
Name	endpoint outage
Mentioned In	n/a

firmware apply

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	firmware apply
Mentioned In	`REQ-COF-004`

firmware artifact

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	firmware artifact
Mentioned In	n/a

firmware bundle

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	firmware bundle
Mentioned In	`Security View`

firmware bundles

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	firmware bundles
Mentioned In	n/a

firmware store

Object store containing signed OTA firmware bundles.

Key	Value
Kind	Catalog Alias
Name	firmware store
Mentioned In	n/a

firmware tampering

Tampered firmware payload attempts to execute on devices.

Key	Value
Kind	Catalog Alias
Name	firmware tampering
Mentioned In	n/a

fleet admin

Owns OTA rollout campaigns and fleet-level reliability.

Key	Value
Kind	Catalog Alias
Name	fleet admin
Mentioned In	n/a

fleet events

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	fleet events
Mentioned In	n/a

fleet operation active

State where machines and central cloud are actively serving operations.

Key	Value
Kind	Catalog Alias
Name	fleet operation active
Mentioned In	n/a

fleet platform operations

Capability area for runtime operations and device identity/secrets lifecycle.

Key	Value
Kind	Catalog Alias
Name	fleet platform operations
Mentioned In	n/a

ingest endpoint unavailable

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Catalog Alias

Key	Value
Name	ingest endpoint unavailable
Mentioned In	n/a

ingestion api

Ingests telemetry and machine status from edge devices.

Key	Value
Kind	Catalog Alias
Name	ingestion api
Mentioned In	n/a

ingestion events

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	ingestion events
Mentioned In	`Architecture Intent View`

iot endpoint

Cloud endpoint for receiving telemetry and status from devices.

Key	Value
Kind	Catalog Alias
Name	iot endpoint
Mentioned In	n/a

logging records

Immutable log record for security and compliance analysis.

Key	Value
Kind	Catalog Alias
Name	logging records
Mentioned In	`REQ-COF-009`

logging service

Managed log and metrics service used for fleet observability.

Key	Value
Kind	Catalog Alias
Name	logging service
Mentioned In	n/a

machine identities

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Catalog Alias
Name	machine identities
Mentioned In	n/a

machine status

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	machine status
Mentioned In	`Architecture Intent View`

machine telemetry

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	machine telemetry
Mentioned In	n/a

mqtt sdk

SDK used by edge components for telemetry and command transport.

Key	Value
Kind	Catalog Alias
Name	mqtt sdk

Key	Value
Mentioned In	n/a

normal mode

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Catalog Alias
Name	normal mode
Mentioned In	n/a

normal operation mode

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Catalog Alias
Name	normal operation mode
Mentioned In	n/a

observability event fanout

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	observability event fanout
Mentioned In	n/a

observability events

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	observability events
Mentioned In	n/a

observability reporting

Aggregates telemetry and update outcomes into operational and audit signals.

Key	Value
Kind	Catalog Alias
Name	observability reporting
Mentioned In	n/a

ota commands

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	ota commands
Mentioned In	n/a

ota enabled

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Catalog Alias
Name	ota enabled
Mentioned In	n/a

ota outcomes

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	ota outcomes
Mentioned In	`Architecture Intent View`

ota rollout

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Catalog Alias
Name	ota rollout
Mentioned In	`Architecture Intent View`, `REQ-COF-003`, `Ota Rollout Flow`

ota rollout commands

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Alias
Name	ota rollout commands
Mentioned In	n/a

ota status reported

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	ota status reported
Mentioned In	n/a

platform ops

Owns runtime deployment and operational controls.

Key	Value
Kind	Catalog Alias
Name	platform ops
Mentioned In	n/a

replay abuse

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Catalog Alias
Name	replay abuse
Mentioned In	`Security View`, `REQ-COF-012`

rollback firmware apply

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias

Key	Value
Name	rollback firmware apply
Mentioned In	`REQ-COF-005`

rollback status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	rollback status
Mentioned In	n/a

rollout cohorts

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Alias
Name	rollout cohorts
Mentioned In	`Architecture Intent View`

rollout windows

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Alias
Name	rollout windows
Mentioned In	n/a

runtime operations

Operates cloud runtime deployment and scaling controls.

Key	Value
Kind	Catalog Alias
Name	runtime operations
Mentioned In	n/a

security reviewer

Reviews identity, firmware integrity, and logging controls.

Key	Value
Kind	Catalog Alias
Name	security reviewer
Mentioned In	n/a

signature valid

Condition indicating firmware signature verification succeeded.

Key	Value
Kind	Catalog Alias
Name	signature valid
Mentioned In	`Security View`, `Ota Campaign Test`

signed firmware bundles

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Alias
Name	signed firmware bundles
Mentioned In	`Architecture Intent View`

signer service

Service that signs firmware artifacts before OTA rollout.

Key	Value
Kind	Catalog Alias
Name	signer service
Mentioned In	n/a

status feedback

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	status feedback

Key	Value
Mentioned In	n/a

store operator

Operates coffee machines and observes local machine health.

Key	Value
Kind	Catalog Alias
Name	store operator
Mentioned In	n/a

telemetry

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	telemetry
Mentioned In	`Architecture Intent View`, `Deployment View`, `Security View`, `Interaction Flow View`, `REQ-COF-001`, `REQ-COF-002`, `REQ-COF-006`, `REQ-COF-008`, `REQ-COF-011`, `Ota Rollout Flow`, `Logging Pipeline Test`, `Telemetry Ingest Contract Test`

telemetry collection

Collects brew/device telemetry and publishes cloud-bound payloads.

Key	Value
Kind	Catalog Alias
Name	telemetry collection
Mentioned In	n/a

telemetry ingested

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Alias
Name	telemetry ingested
Mentioned In	n/a

telemetry payloads

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	telemetry payloads
Mentioned In	`Architecture Intent View`

telemetry records

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Alias
Name	telemetry records
Mentioned In	n/a

telemetry replay

Replayed telemetry floods ingestion and distorts reporting.

Key	Value
Kind	Catalog Alias
Name	telemetry replay
Mentioned In	n/a

telemetry replay detected

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Catalog Alias
Name	telemetry replay detected
Mentioned In	n/a

the coffee fleet system

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture.

Key	Value
Kind	Catalog Alias
Name	the coffee fleet system

Key	Value
Mentioned In	`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-003`, `REQ-COF-004`, `REQ-COF-005`, `REQ-COF-006`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-011`, `REQ-COF-012`

update outcomes

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	update outcomes
Mentioned In	n/a

update status

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Alias
Name	update status
Mentioned In	n/a

update/rollback status

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Alias
Name	update/rollback status
Mentioned In	n/a

validation failed

Event raised when signature/hash validation fails on firmware.

Key	Value
Kind	Catalog Alias
Name	validation failed
Mentioned In	n/a

AV-LOG-TAMPERING

Attempts to suppress or alter security/operational log records.

Key	Value
Kind	Catalog Attack Vector
Name	log tampering
Mentioned In	`Security View`

AV-MALICIOUS-FIRMWARE-BUNDLE

Tampered firmware payload attempts to execute on devices.

Key	Value
Kind	Catalog Attack Vector
Name	malicious firmware bundle
Mentioned In	`Security View`

AV-SPOOFED-DEVICE-IDENTITY

Unauthorized actor impersonates a machine identity.

Key	Value
Kind	Catalog Attack Vector
Name	spoofed device identity
Mentioned In	`Security View`

AV-TELEMETRY-REPLAY-ABUSE

Replayed telemetry floods ingestion and distorts reporting.

Key	Value
Kind	Catalog Attack Vector
Name	telemetry replay abuse
Mentioned In	`Security View`

COND-FIRMWARE-SIGNATURE-VALID

Condition indicating firmware signature verification succeeded.

Key	Value
Kind	Catalog Condition
Name	firmware signature is valid

Key	Value
Mentioned In	`REQ-COF-004`

DATA-AUDIT-LOG-RECORD

Immutable log record for security and compliance analysis.

Key	Value
Kind	Catalog Data Term
Name	audit log record
Mentioned In	`Interaction Flow View`, `REQ-COF-007`

DATA-BREW-TELEMETRY

Machine telemetry including brew duration, temperature, and status metadata.

Key	Value
Kind	Catalog Data Term
Name	brew telemetry
Mentioned In	`REQ-COF-001`

DATA-CAMPAIGN-ID

Stable OTA campaign identifier used to correlate rollout and audit records.

Key	Value
Kind	Catalog Data Term
Name	campaign id
Mentioned In	`REQ-COF-007`

DATA-FIRMWARE-BUNDLE

Signed OTA firmware artifact distributed to machine cohorts for update apply.

Key	Value
Kind	Catalog Data Term
Name	signed firmware bundle
Mentioned In	`REQ-COF-003`

DATA-MACHINE-IDENTITY

Device identity material used for authenticated publish, ingest, and audit attribution.

Key	Value
Kind	Catalog Data Term
Name	machine identity
Mentioned In	`Interaction Flow View`, `REQ-COF-007`

DATA-OTA-CAMPAIGN-POLICY

Policy describing rollout cohorts, windows, and rollback criteria.

Key	Value
Kind	Catalog Data Term
Name	ota campaign policy
Mentioned In	`REQ-COF-003`

DATA-OTA-STATUS

Device-level status for firmware download, apply, and rollback outcomes.

Key	Value
Kind	Catalog Data Term
Name	ota status
Mentioned In	`REQ-COF-010`

EVT-BREW-CYCLE-COMPLETED

Event raised when a coffee machine completes a brew cycle.

Key	Value
Kind	Catalog Event
Name	brew cycle completed event is received
Mentioned In	`REQ-COF-001`

EVT-CLOUD-INGEST-ENDPOINT-UNAVAILABLE

Event raised when cloud ingest endpoint becomes unavailable for telemetry ingestion.

Key	Value
Kind	Catalog Event
Name	cloud ingest endpoint is unavailable
Mentioned In	`REQ-COF-006`, `REQ-COF-011`

EVT-FIRMWARE-VALIDATION-FAILED

Event raised when signature/hash validation fails on firmware.

Key	Value
Kind	Catalog Event
Name	firmware validation failed event is received
Mentioned In	`REQ-COF-005`, `REQ-COF-010`

EVT-OTA-CAMPAIGN-SCHEDULED

Event raised when a fleet OTA campaign is scheduled.

Key	Value
Kind	Catalog Event
Name	ota campaign scheduled event is received
Mentioned In	`REQ-COF-003`, `REQ-COF-004`

EVT-OTA-STATUS-REPORTED

Event raised when device reports OTA apply status.

Key	Value
Kind	Catalog Event
Name	ota status reported event is received
Mentioned In	`REQ-COF-007`

EVT-TELEMETRY-INGESTED

Event raised when cloud ingestion accepts telemetry.

Key	Value
Kind	Catalog Event
Name	telemetry ingested event is received
Mentioned In	`REQ-COF-002`, `REQ-COF-009`

EVT-TELEMETRY-REPLAY-ABUSE-DETECTED

Event raised when replayed telemetry behavior is detected and flagged as abuse.

Key	Value
Kind	Catalog Event

Key	Value
Name	telemetry replay abuse is detected
Mentioned In	`REQ-COF-008`, `REQ-COF-012`

FEAT-AUDIT-LOGGING

Feature flag enabling audit-grade log persistence.

Key	Value
Kind	Catalog Feature
Name	audit logging is enabled
Mentioned In	`REQ-COF-007`

FEAT-OTA-CAMPAIGN

Feature flag enabling centralized OTA rollout behavior.

Key	Value
Kind	Catalog Feature
Name	ota campaign is enabled
Mentioned In	`REQ-COF-003`, `REQ-COF-004`

FG-CLOUD-CONTROL

Capability area for fleet ingestion, campaign orchestration, and cloud reporting.

Key	Value
Kind	Catalog Functional Group
Name	cloud control
Mentioned In	`Architecture Intent View`, `Deployment View`, `Security View`, `REQ-COF-003`

FG-MACHINE-EDGE

Capability area for device-local telemetry and OTA execution behavior.

Key	Value
Kind	Catalog Functional Group
Name	machine edge
Mentioned In	`Architecture Intent View`, `Deployment View`, `Security View`

FG-PLATFORM-OPERATIONS

Capability area for runtime operations and device identity/secrets lifecycle.

Key	Value
Kind	Catalog Functional Group
Name	platform operations
Mentioned In	`Architecture Intent View`, `Deployment View`, `Security View`

FU-CLOUD-RUNTIME-OPERATIONS

Operates cloud runtime deployment and scaling controls.

Key	Value
Kind	Catalog Functional Unit
Name	cloud runtime operations
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

FU-DEVICE-IDENTITY-SECRETS

Manages device identity material and secure configuration.

Key	Value
Kind	Catalog Functional Unit
Name	device identity and secrets
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Logging Pipeline Test`, `Ota Campaign Test`

FU-FLEET-INGESTION-API

Ingests telemetry and machine status from edge devices.

Key	Value
Kind	Catalog Functional Unit
Name	fleet ingestion api
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Interaction Flow View`, `REQ-COF-001`, `Logging Pipeline Test`, `Telemetry Ingest Contract Test`

FU-FLEET-OBSERVABILITY-REPORTING

Aggregates telemetry and update outcomes into operational and audit signals.

Key	Value
Kind	Catalog Functional Unit

Key	Value
Name	fleet observability reporting
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Interaction Flow View`, `Ota Rollout Flow`, `Logging Pipeline Test`

FU-MACHINE-TELEMETRY-COLLECTION

Collects brew/device telemetry and publishes cloud-bound payloads.

Key	Value
Kind	Catalog Functional Unit
Name	machine telemetry collection
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Interaction Flow View`, `Telemetry Ingest Contract Test`

FU-OTA-UPDATE-AGENT

Downloads, validates, and applies signed firmware updates on machines.

Key	Value
Kind	Catalog Functional Unit
Name	ota update agent
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Interaction Flow View`, `Ota Rollout Flow`, `Ota Campaign Test`

FU-UPDATE-CAMPAIGN-ORCHESTRATION

Selects target machine groups and coordinates OTA rollout phases.

Key	Value
Kind	Catalog Functional Unit
Name	update campaign orchestration
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`, `Interaction Flow View`, `Ota Rollout Flow`, `Ota Campaign Test`

MODE-NORMAL

Default operation mode for telemetry and OTA behavior.

Key	Value
Kind	Catalog Mode

Key	Value
Name	normal mode is active
Mentioned In	`REQ-COF-011`

REF-CLOUD-LOGGING-SERVICE

Managed log and metrics service used for fleet observability.

Key	Value
Kind	Catalog Referenced Element
Name	cloud logging service
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

REF-CLOUD-RUNTIME-SDK

SDK used by cloud units for runtime integrations.

Key	Value
Kind	Catalog Referenced Element
Name	cloud runtime sdk
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

REF-FIRMWARE-BUNDLE-STORE

Object store containing signed OTA firmware bundles.

Key	Value
Kind	Catalog Referenced Element
Name	firmware bundle store
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

REF-FIRMWARE-SIGNER-SERVICE

Service that signs firmware artifacts before OTA rollout.

Key	Value
Kind	Catalog Referenced Element
Name	firmware signer service
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

REF-IOT-INGEST-ENDPOINT

Cloud endpoint for receiving telemetry and status from devices.

Key	Value
Kind	Catalog Referenced Element
Name	iot ingest endpoint
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

REF-MQTT-DEVICE-SDK

SDK used by edge components for telemetry and command transport.

Key	Value
Kind	Catalog Referenced Element
Name	mqtt device sdk
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Security View`, `Traceability View`

STATE-FLEET-OPERATION-ACTIVE

State where machines and central cloud are actively serving operations.

Key	Value
Kind	Catalog State
Name	coffee machine fleet operation is active
Mentioned In	`REQ-COF-001`, `REQ-COF-002`, `REQ-COF-009`

SYS-COFFEE-FLEET-SYSTEM

Connected coffee-machine fleet and central cloud control system-of-interest covered by this architecture.

Key	Value
Kind	Catalog System
Name	coffee fleet system
Mentioned In	n/a

EM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor

Key	Value
Mentioned In	`Architecture Intent View`, `Communication View`, `Interaction Flow View`

EM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

EM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term
Name	authored mapping
Mentioned In	`Communication View`, `Security View`

EM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element
Mentioned In	n/a

EM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

Key	Value
Kind	Engineering Model Term
Name	control
Mentioned In	`Architecture Intent View`, `Deployment View`, `Security View`, `State Lifecycle View`, `REQ-COF-004`, `REQ-COF-008`

EM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

Key	Value
Kind	Engineering Model Term
Name	data object
Mentioned In	n/a

EM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

Key	Value
Kind	Engineering Model Term
Name	deployment target
Mentioned In	n/a

EM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Key	Value
Kind	Engineering Model Term
Name	event
Mentioned In	`Architecture Intent View`, `Communication View`, `Security View`, `State Lifecycle View`, `REQ-COF-012`

EM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

Key	Value
Kind	Engineering Model Term
Name	flow
Mentioned In	`Architecture Intent View`, `Communication View`, `Interaction Flow View`

EM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

Key	Value
Kind	Engineering Model Term
Name	flow step

Key	Value
Mentioned In	n/a

EM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term
Name	functional group
Mentioned In	n/a

EM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term
Name	functional unit
Mentioned In	`State Lifecycle View`

EM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

EM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Key	Value
Kind	Engineering Model Term
Name	interface
Mentioned In	`Architecture Intent View`, `Communication View`, `Deployment View`, `Interaction Flow View`

EM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

EM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	`Traceability View`, `Interaction Flow View`, `REQ-COF-002`, `REQ-COF-007`, `REQ-COF-008`, `REQ-COF-009`, `REQ-COF-010`, `REQ-COF-012`

EM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

EM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Key	Value
Kind	Engineering Model Term
Name	state
Mentioned In	`Architecture Intent View`, `State Lifecycle View`

EM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

Key	Value
Kind	Engineering Model Term
Name	trust boundary

Key	Value
Mentioned In	`Architecture Intent View`, `Security View`, `REQ-COF-004`

EM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

EM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term
Name	verification check
Mentioned In	n/a

13.3. Inferred Runtime References

ota_campaign_schedule

schedules OTA campaign orchestration triggers for configured rollout windows

Key	Value
Kind	Runtime eventbridge_rule
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Deployment View`

ota_campaign_target

routes scheduled OTA campaign events to the orchestration runtime handler

Key	Value
Kind	Runtime eventbridge_target
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

Key	Value
Mentioned In	`Deployment View`

fleet_ingestion_api

receives device telemetry, validates payload shape, and publishes normalized fleet events

Key	Value
Kind	Runtime lambda_function
Owner	FU-FLEET-INGESTION-API
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Deployment View`, `Security View`

fleet_observability_reporting

aggregates telemetry outcomes into fleet operations and security reporting signals

Key	Value
Kind	Runtime lambda_function
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Deployment View`, `Security View`

update_campaign_orchestration

coordinates OTA campaign rollout windows and device cohort targeting

Key	Value
Kind	Runtime lambda_function
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Deployment View`, `Security View`

telemetry_retry_queue

buffers telemetry during ingest outages for retry once normal mode resumes

Key	Value
Kind	Runtime queue
Owner	FU-CLOUD-RUNTIME-OPERATIONS

Key	Value
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Deployment View`

fleet_event_topic

broadcasts fleet-wide telemetry and OTA lifecycle events to subscribed processors

Key	Value
Kind	Runtime topic
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf
Mentioned In	`Deployment View`

13.4. Inferred Code References

cloud_runtime_ops.go

wires cloud runtime handlers and integration entrypoints

Key	Value
Kind	Code source_file
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	cloud_runtime_ops.go
Mentioned In	`Deployment View`

device_identity_secrets.ts

resolves device identity credentials and firmware verification secrets

Key	Value
Kind	Code source_file
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts
Mentioned In	`Deployment View`, `Security View`

fleet_ingestion_api.go

ingests device telemetry and emits normalized fleet ingestion events

Key	Value
Kind	Code source_file
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go
Mentioned In	n/a

fleet_observability_reporting.ts

aggregates telemetry and OTA outcome signals for fleet observability reporting

Key	Value
Kind	Code source_file
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts
Mentioned In	n/a

machine_telemetry.rs

builds machine telemetry and status payloads at the edge

Key	Value
Kind	Code source_file
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs
Mentioned In	`Deployment View`, `Security View`

ota_update_agent.rs

applies OTA campaign updates and reports rollout outcomes

Key	Value
Kind	Code source_file
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs
Mentioned In	`Deployment View`, `Security View`

update_campaign_orchestrator.ts

plans OTA rollout cohorts and campaign lifecycle transitions

Key	Value
Kind	Code source_file
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts
Mentioned In	`Deployment View`, `Security View`

CODE-HANDLEROLLBACK

Inferred code symbol owned by FU-UPDATE-CAMPAIGN-ORCHESTRATION from update_campaign_orchestrator.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts:10
Mentioned In	n/a

CODE-INGESTTELEMETRY

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:11.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:11
Mentioned In	n/a

CODE-NOTIFYREPLAYABUSE

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:15.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:15
Mentioned In	n/a

CODE-PERSISTAUDITLOG

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:10
Mentioned In	n/a

CODE-PERSISTTELEMETRYMETRIC

Inferred code symbol owned by FU-FLEET-OBSERVABILITY-REPORTING from fleet_observability_reporting.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-OBSERVABILITY-REPORTING
Source	fleet_observability_reporting.ts:5
Mentioned In	n/a

CODE-PLANCAMPAIGN

Inferred code symbol owned by FU-UPDATE-CAMPAIGN-ORCHESTRATION from update_campaign_orchestrator.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-UPDATE-CAMPAIGN-ORCHESTRATION
Source	update_campaign_orchestrator.ts:5
Mentioned In	n/a

CODE-PUBLISHBREWTELEMETRY

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:10.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:10
Mentioned In	n/a

CODE-QUEUEFORRETRY

Inferred code symbol owned by FU-MACHINE-TELEMETRY-COLLECTION from machine_telemetry.rs:15.

Key	Value
Kind	Code symbol
Owner	FU-MACHINE-TELEMETRY-COLLECTION
Source	machine_telemetry.rs:15
Mentioned In	n/a

CODE-RAISESECURITYNOTIFICATION

Inferred code symbol owned by FU-CLOUD-RUNTIME-OPERATIONS from cloud_runtime_ops.go:6.

Key	Value
Kind	Code symbol
Owner	FU-CLOUD-RUNTIME-OPERATIONS
Source	cloud_runtime_ops.go:6
Mentioned In	n/a

CODE-REDACTIDENTITYFORAUDIT

Inferred code symbol owned by FU-DEVICE-IDENTITY-SECRETS from device_identity_secrets.ts:10.

Key	Value
Kind	Code symbol
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts:10
Mentioned In	n/a

CODE-REJECTREPLAY

Inferred code symbol owned by FU-FLEET-INGESTION-API from fleet_ingestion_api.go:16.

Key	Value
Kind	Code symbol
Owner	FU-FLEET-INGESTION-API
Source	fleet_ingestion_api.go:16
Mentioned In	n/a

CODE-ROLLBACKANDREPORT

Inferred code symbol owned by FU-OTA-UPDATE-AGENT from ota_update_agent.rs:14.

Key	Value
Kind	Code symbol
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs:14
Mentioned In	n/a

CODE-VALIDATEANDAPPLY

Inferred code symbol owned by FU-OTA-UPDATE-AGENT from ota_update_agent.rs:5.

Key	Value
Kind	Code symbol
Owner	FU-OTA-UPDATE-AGENT
Source	ota_update_agent.rs:5
Mentioned In	n/a

CODE-VERIFYFIRMWARESIGNATURE

Inferred code symbol owned by FU-DEVICE-IDENTITY-SECRETS from device_identity_secrets.ts:5.

Key	Value
Kind	Code symbol
Owner	FU-DEVICE-IDENTITY-SECRETS
Source	device_identity_secrets.ts:5
Mentioned In	n/a

13.5. Verification References

VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC

runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

Key	Value
Kind	e2e
Status	partial
Source	test-results/ota-rollback-report.md (+1)
Name	Ota Rollout Flow
Mentioned In	`Interaction Flow View`, `Ota Rollout Flow`

VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF

validates telemetry logging pipeline persistence and operational reporting signals

Key	Value
Kind	integration
Status	partial
Source	test-results/integration-logging.json (+1)
Name	Logging Pipeline Test
Mentioned In	`Interaction Flow View`, `Logging Pipeline Test`

VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794

validates telemetry ingest contract fields and accepted request semantics

Key	Value
Kind	test
Status	pass
Source	test-results/contract-telemetry.json (+1)
Name	Telemetry Ingest Contract Test
Mentioned In	`Interaction Flow View`, `Telemetry Ingest Contract Test`

VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8

verifies OTA campaign signature validation and rollout trigger behavior

Key	Value
Kind	unit
Status	pass
Source	test-results/unit-ota.xml (+1)
Name	Ota Campaign Test
Mentioned In	`Interaction Flow View`, `Ota Campaign Test`